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Schuster et al.

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(54) **PROJECTION EXPOSURE SYSTEM**

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(30) **Foreign Application Priority Data**

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(51) **Int. Cl.**[7] **G03B 27/54**; G02B 3/00; G02B 9/00

(52) **U.S. Cl.** **355/67**; 359/349; 359/754

(58) **Field of Search** 355/67; 359/349, 359/359, 754

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(74) *Attorney, Agent, or Firm*—Townsend and Townsend and Crew LLP

(57) **ABSTRACT**

A projection exposure system is proposed which is positionable between a first object and a second object for imaging the first object in a region of the second object with light of a wavelength band having a width $\delta\lambda$ about a central working wavelength λ, wherein a relative width $\delta\lambda/\lambda$ of the wavelength band is larger than 0.002, in particular, larger than 0.005, for example, of the Hg-I-line. The projection exposure system is a so-called three-bulge system comprising three bulges having, as a whole, a positive refractive power and two waists having, as a whole, a negative refractive power. By applying suitable measures, in particular, by suitably selecting the material for the lenses forming the projection exposure system, the long-term stability of the system is increased.

39 Claims, 18 Drawing Sheets

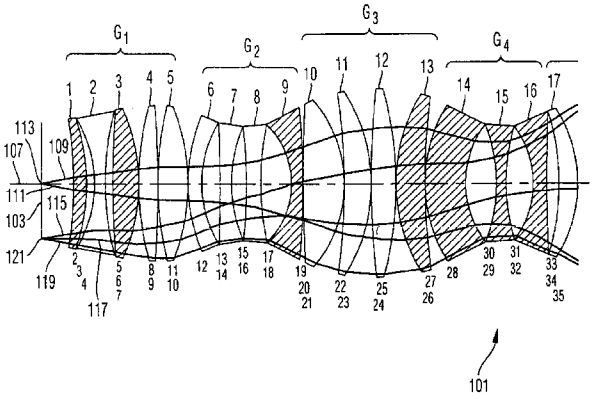


Fig. 1

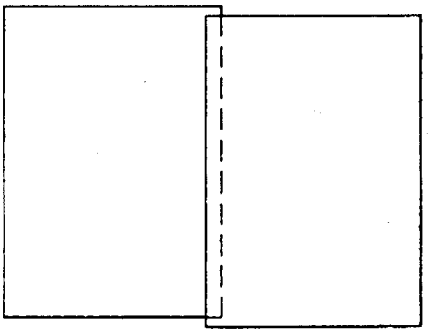


Fig. 1a Fig. 1b

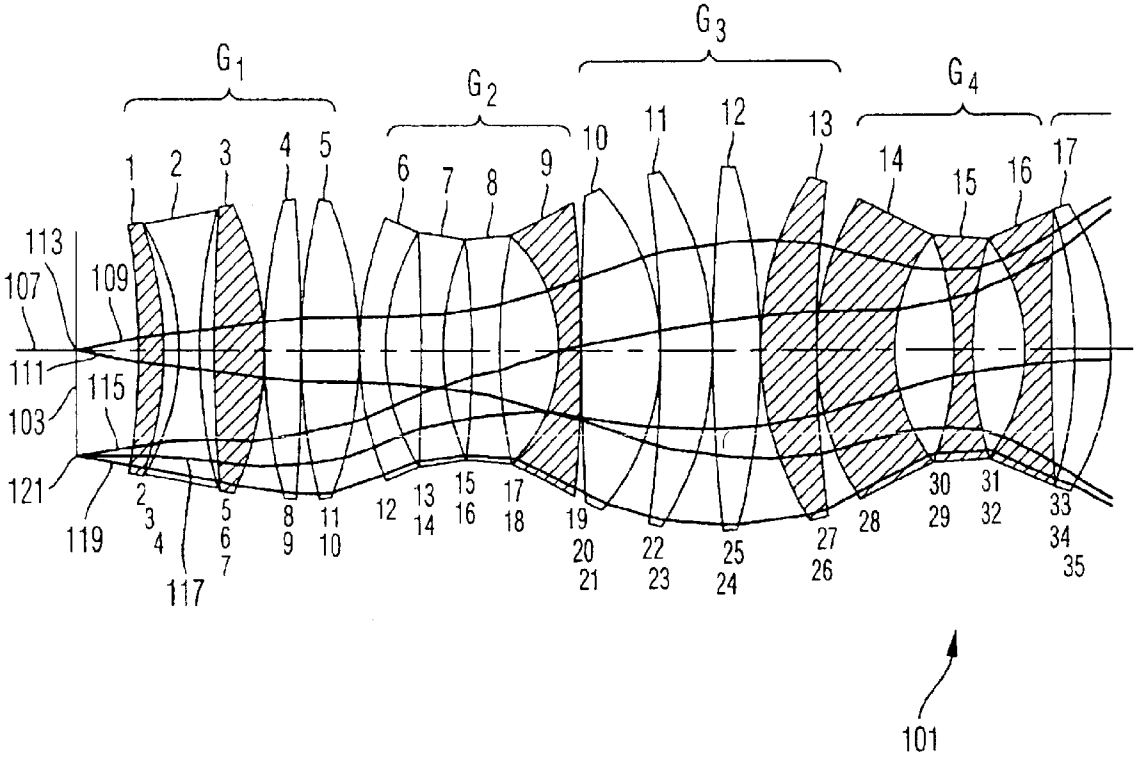


Fig. 1a

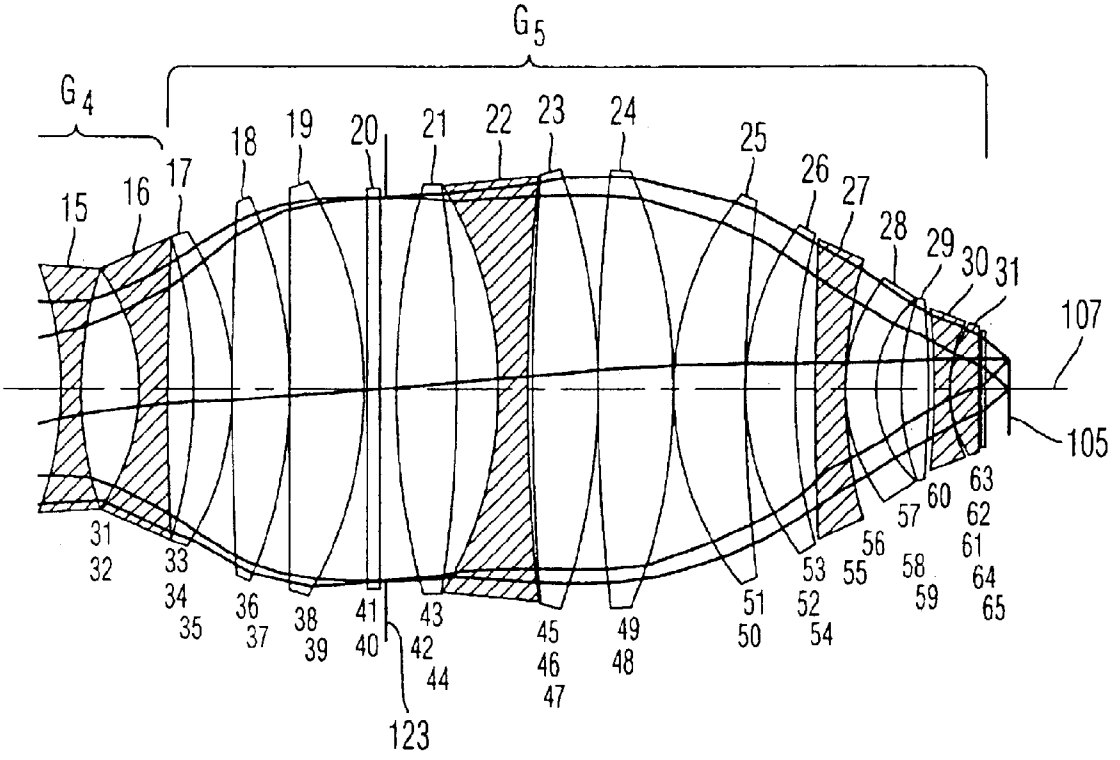

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Fig. 1b

Fig. 2

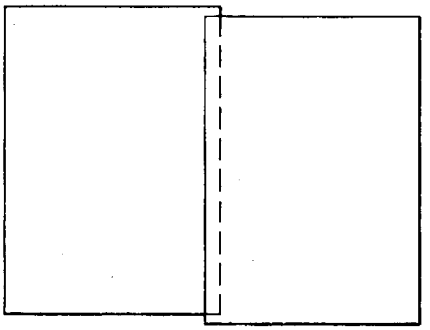


Fig. 2a Fig. 2b

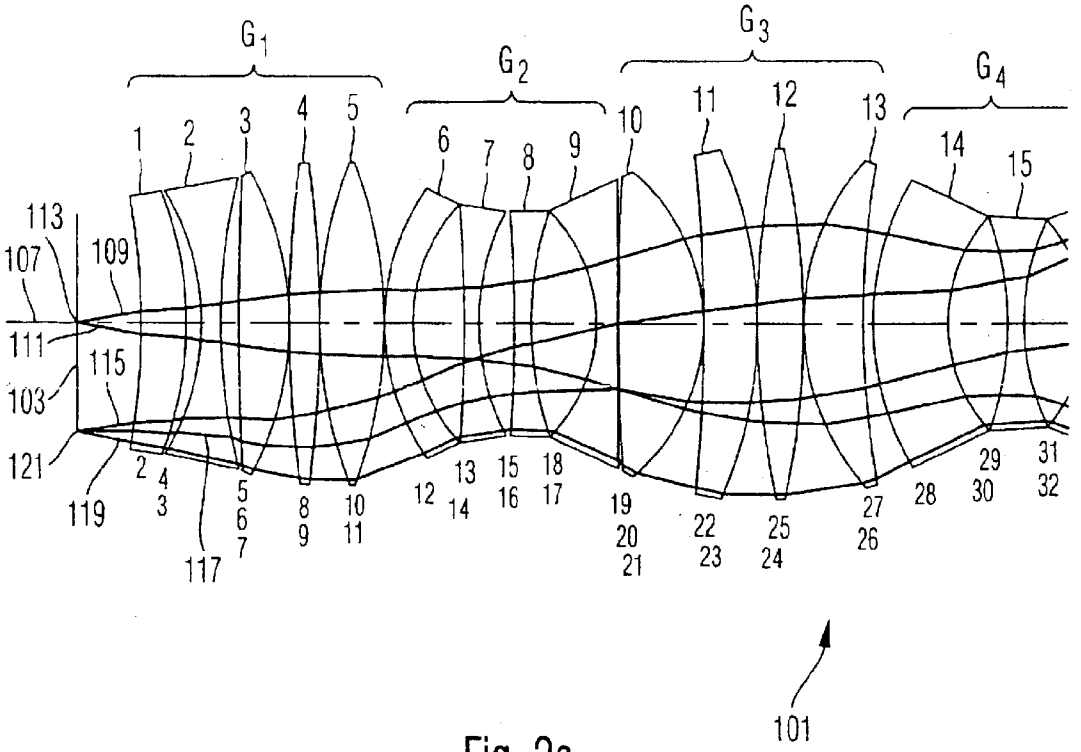


Fig. 2a

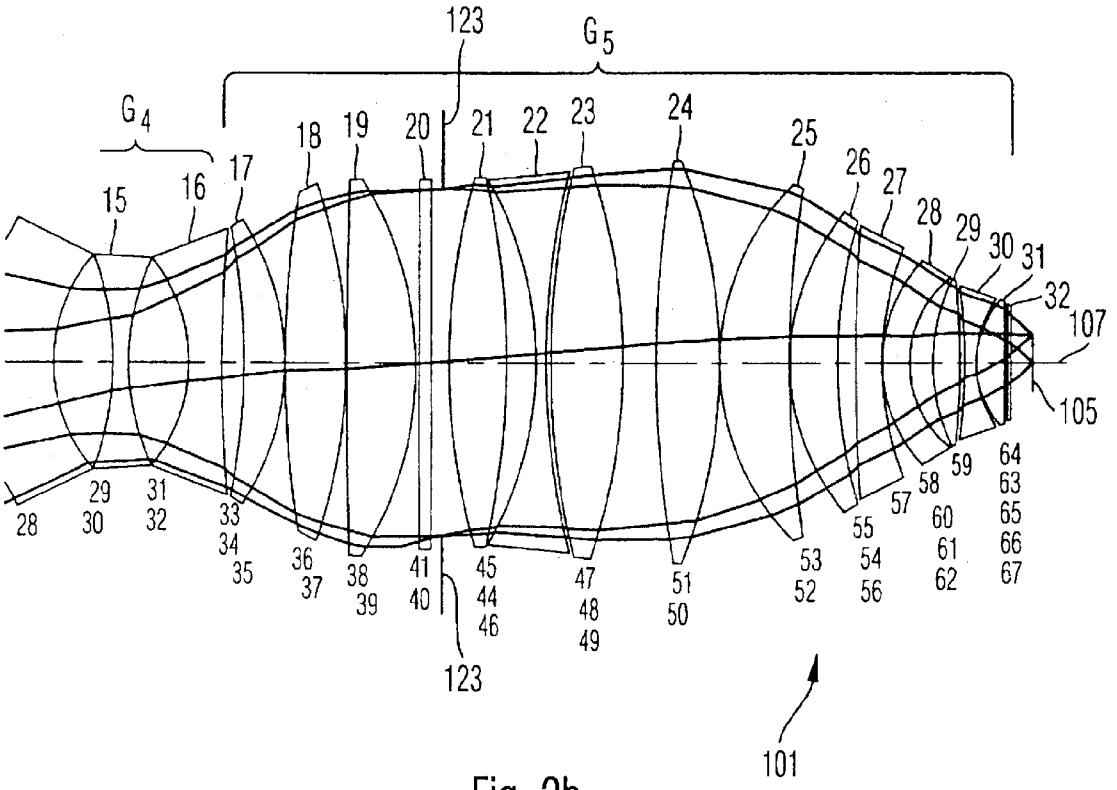

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Fig. 2b

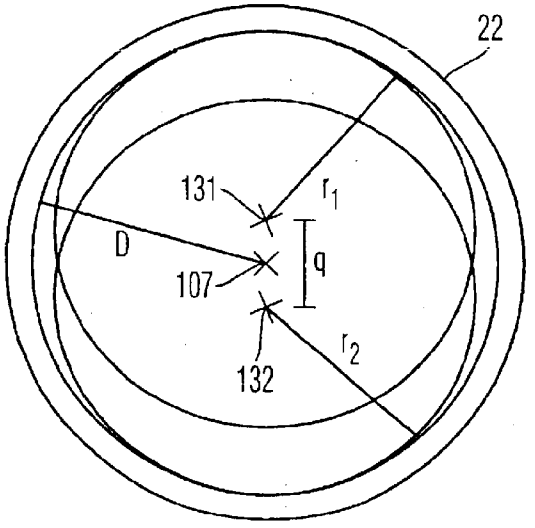

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Fig. 3

Fig. 4

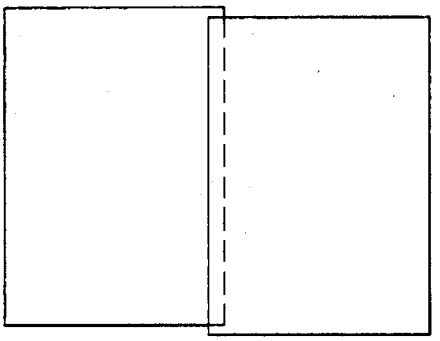


Fig. 4a Fig. 4b

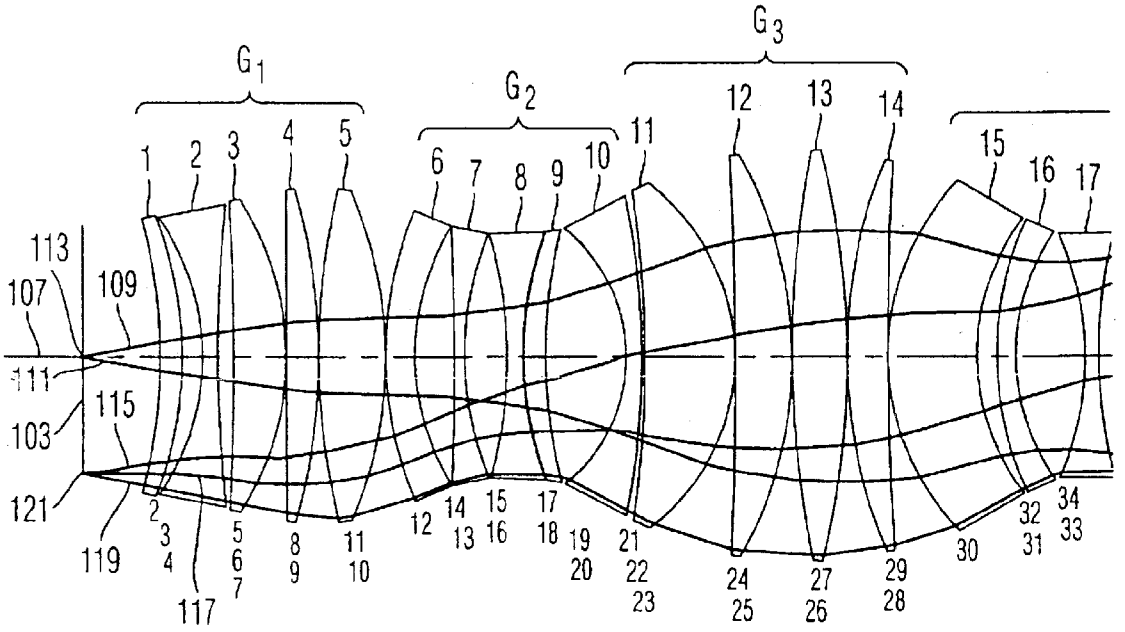


Fig. 4a

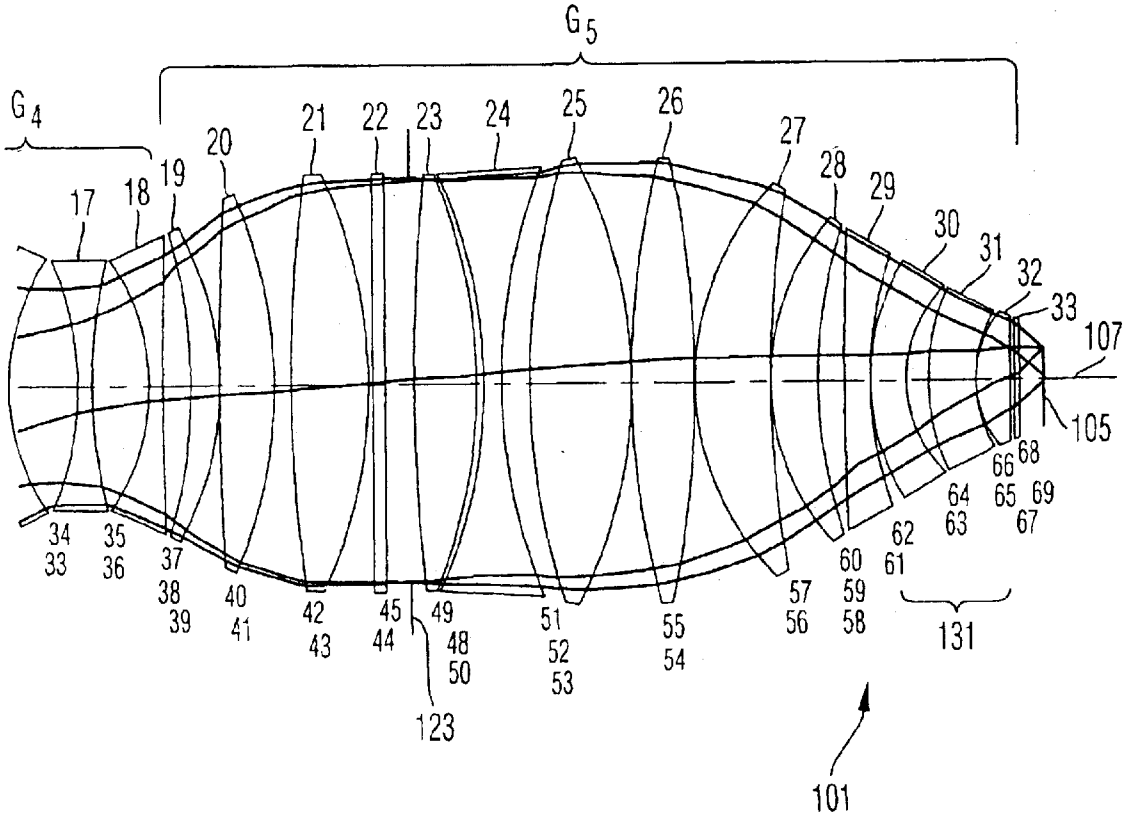

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Fig. 4b

Fig. 5

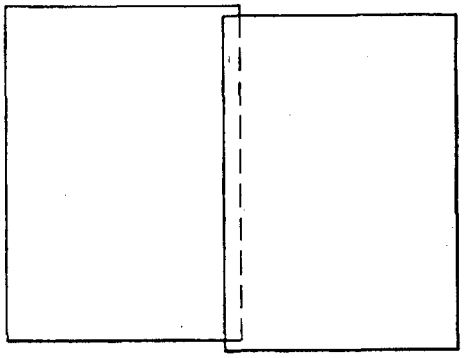


Fig. 5a Fig. 5b

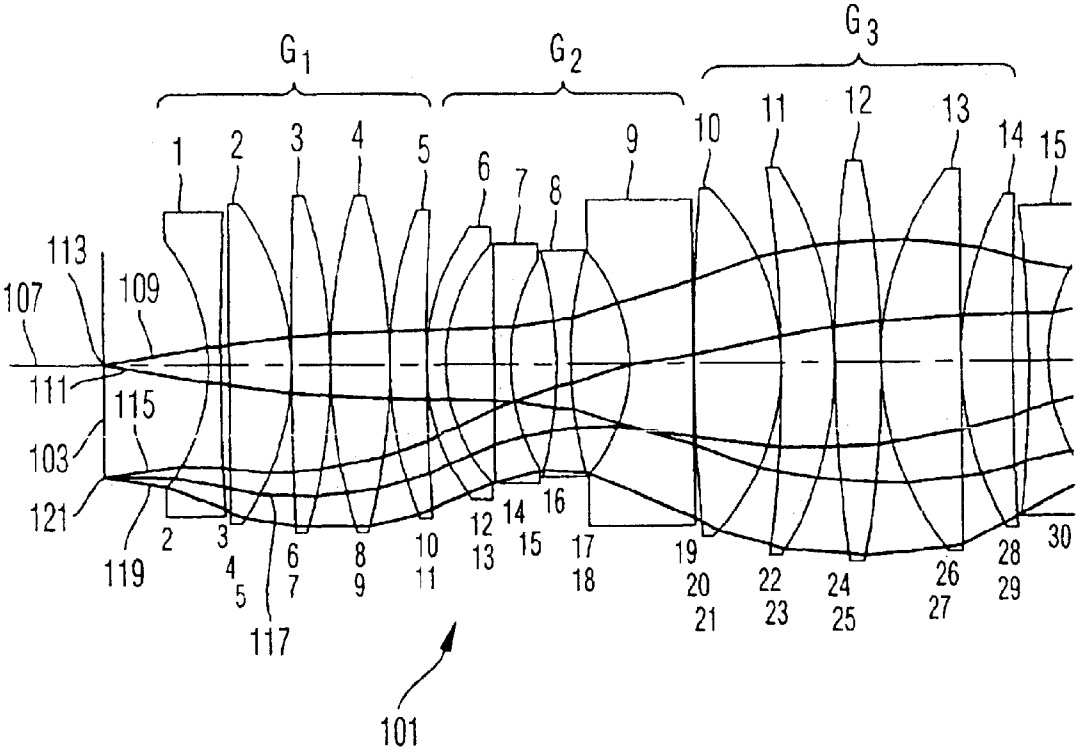


Fig. 5a

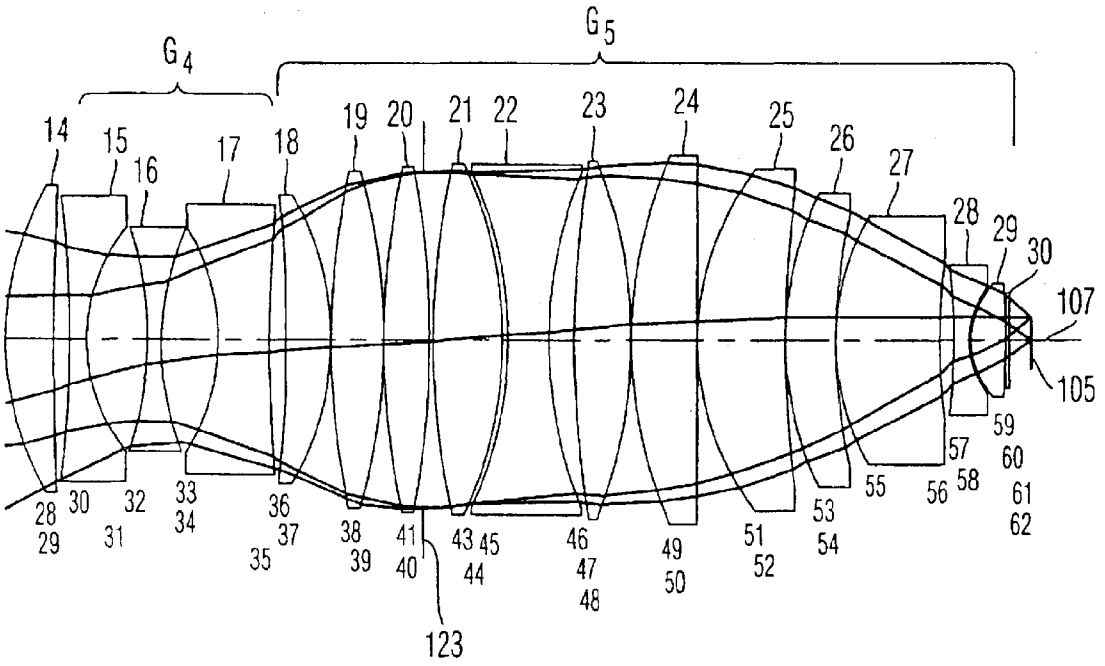

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Fig. 5b

Fig. 6

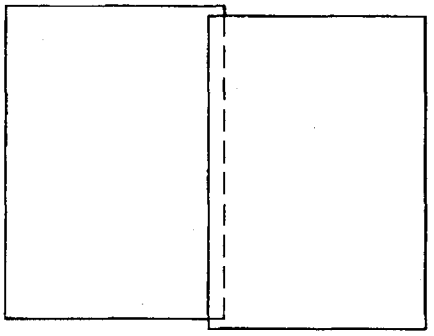


Fig. 6a Fig. 6b

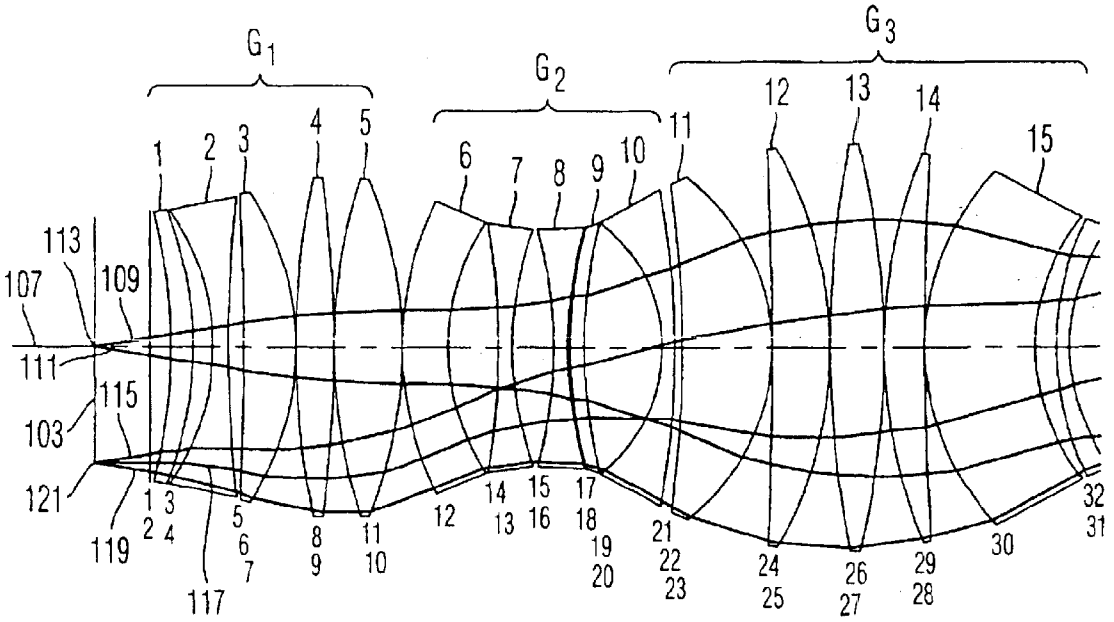


Fig. 6a

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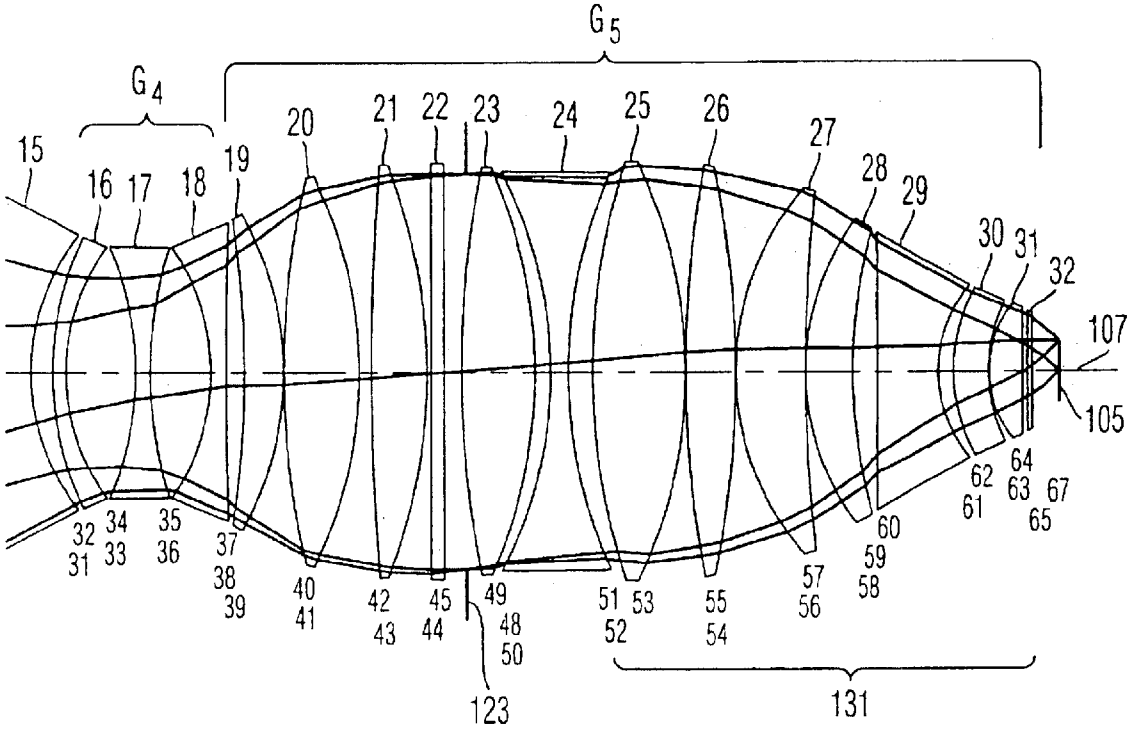

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Fig. 6b

Fig. 7

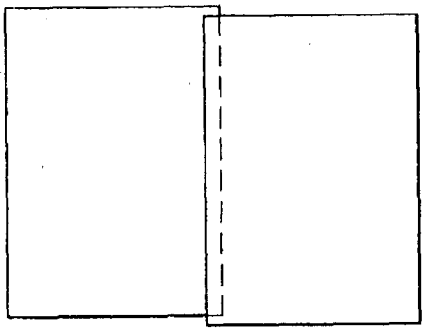


Fig. 7a Fig. 7b

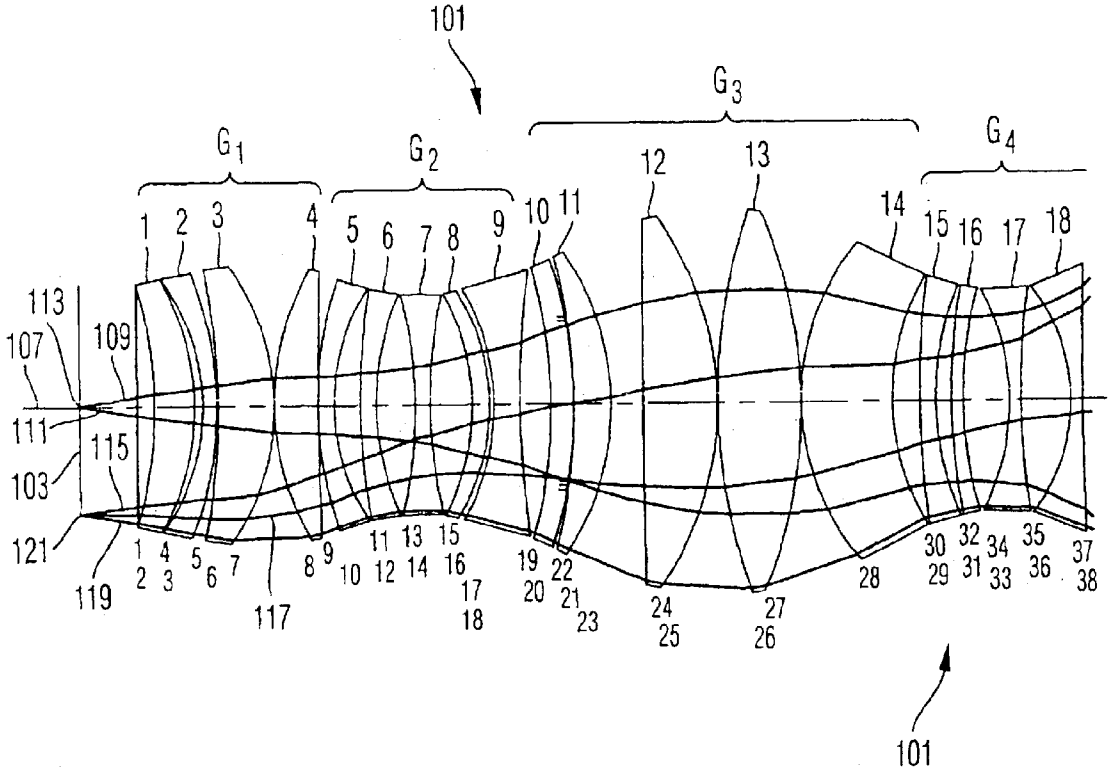


Fig. 7a

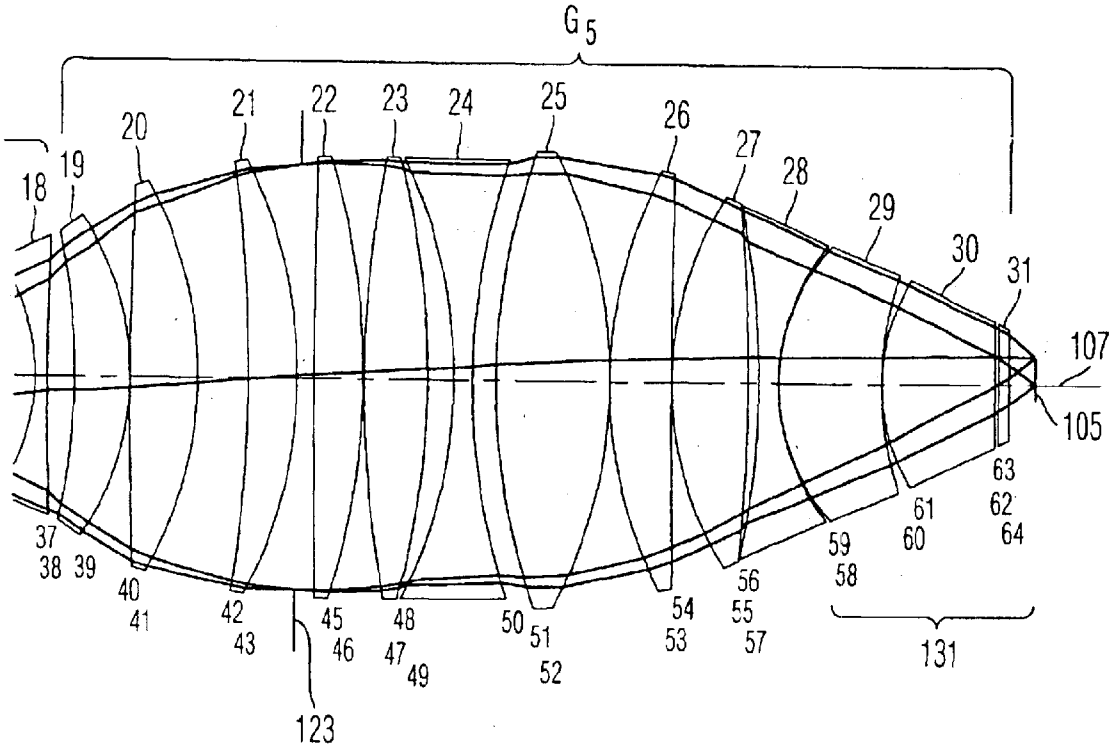

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Fig. 7b

Fig. 8

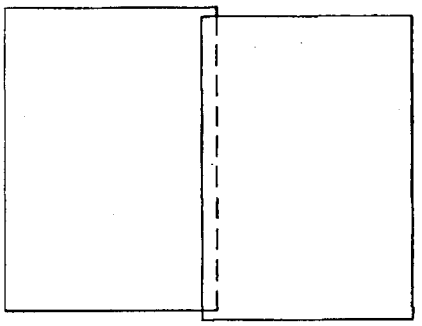


Fig. 8a Fig. 8b

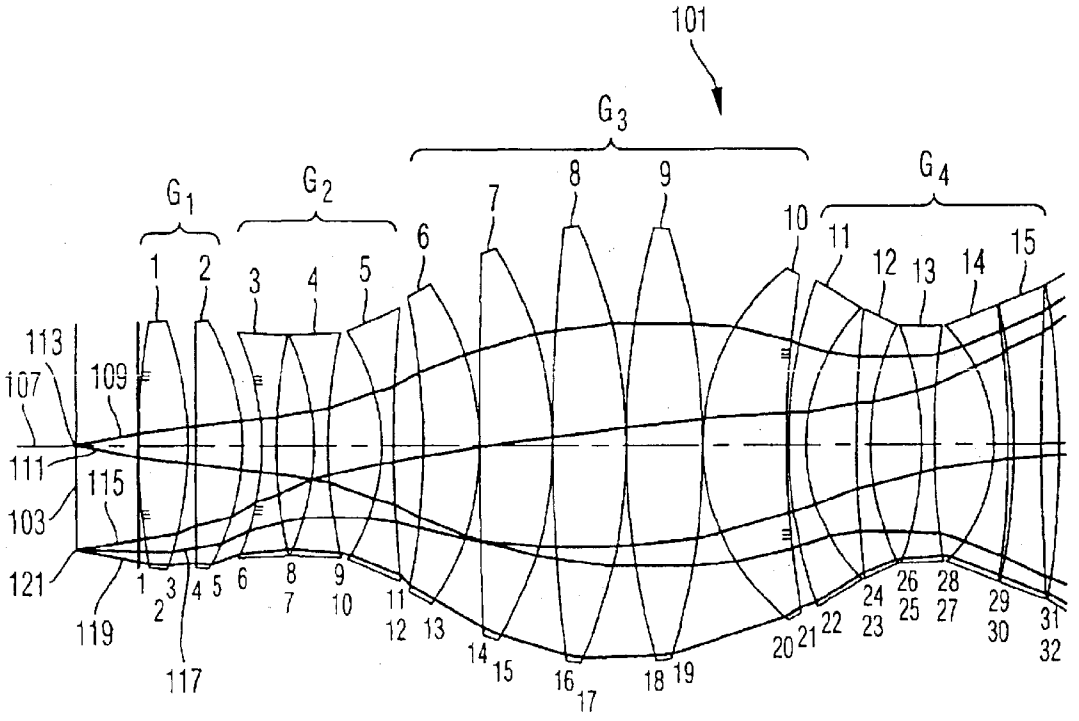


Fig. 8a

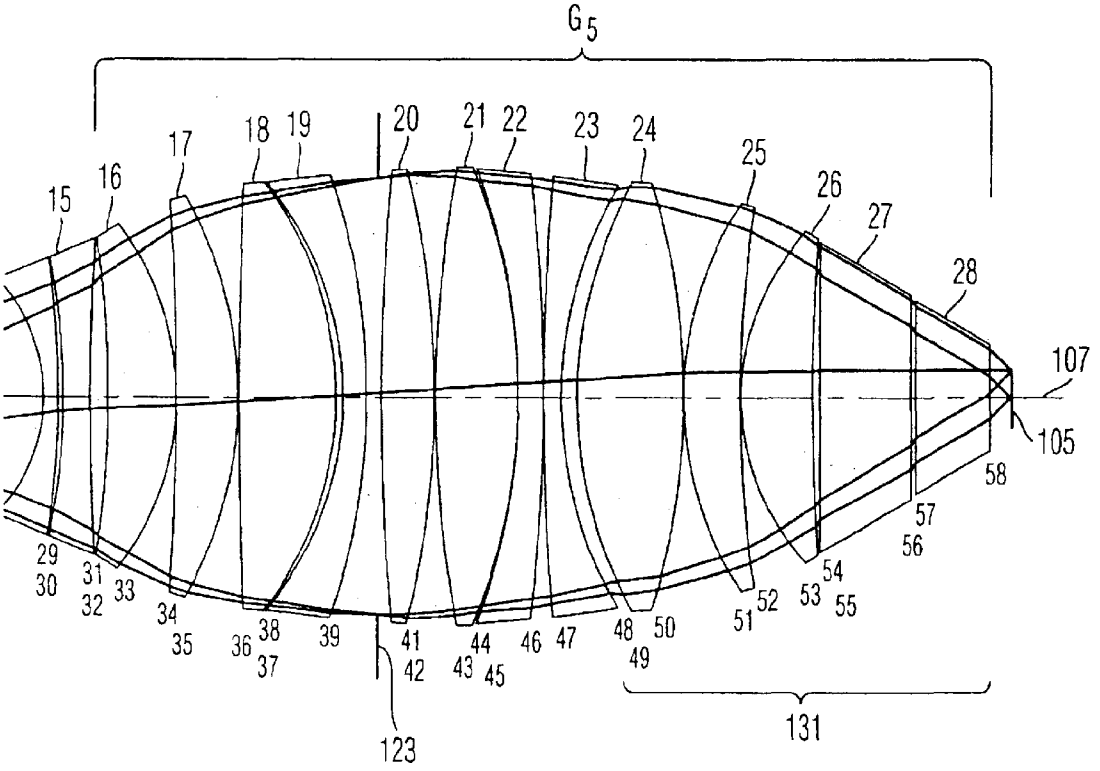

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Fig. 8b

Fig. 9

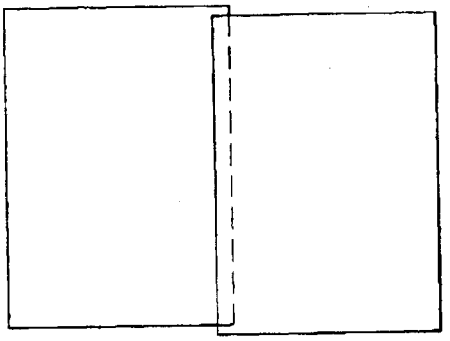


Fig. 9a Fig. 9b

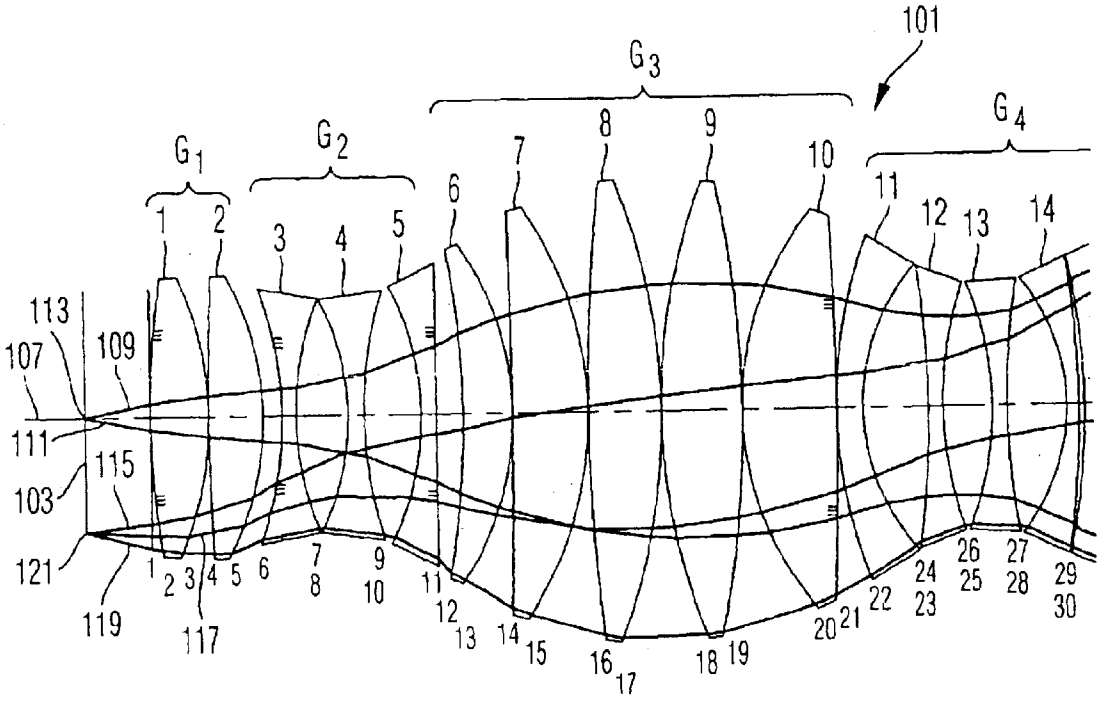


Fig. 9a

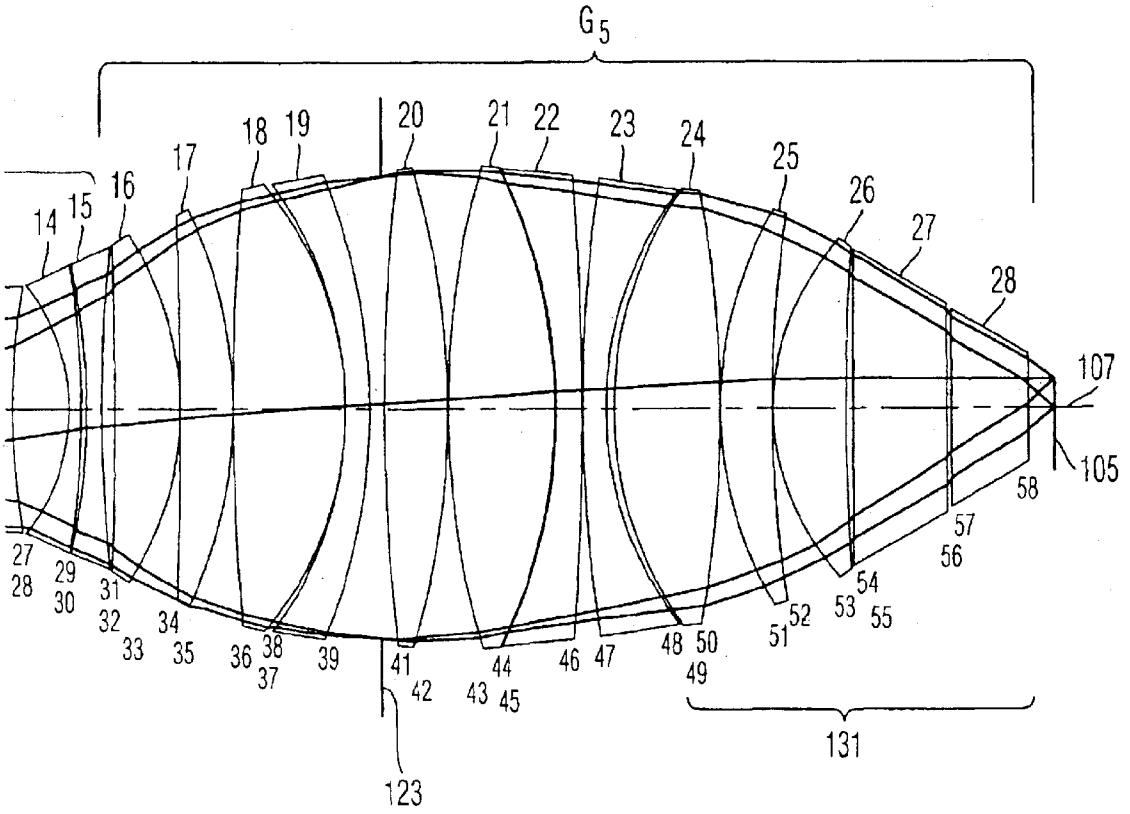

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Fig. 9b

Fig. 10

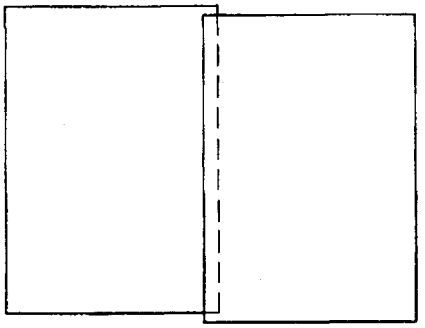


Fig. 10a Fig. 10b

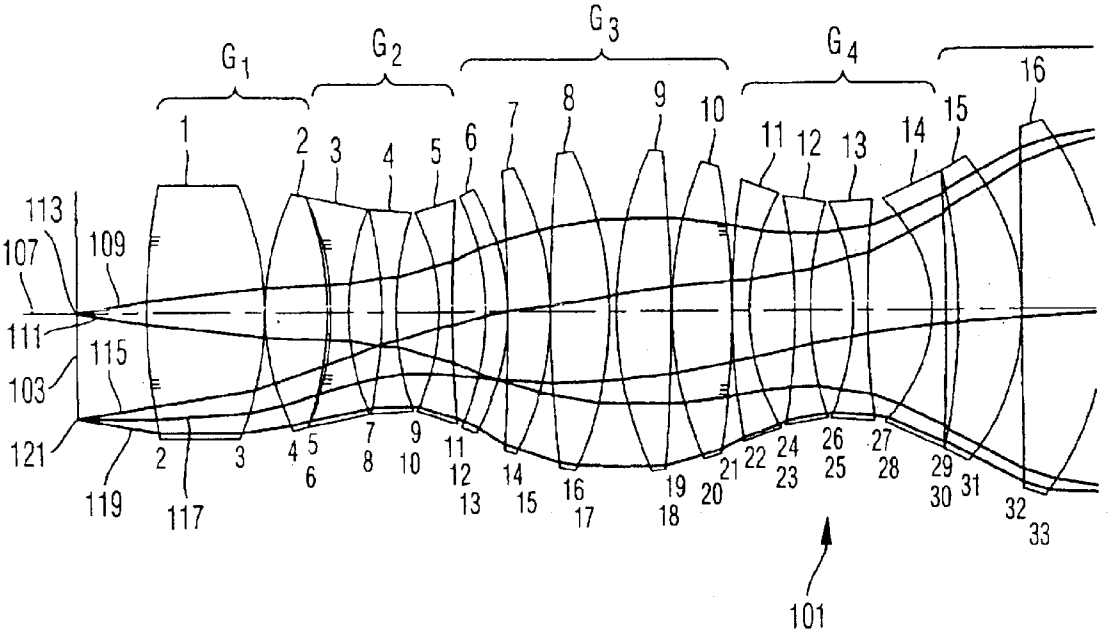


Fig. 10a

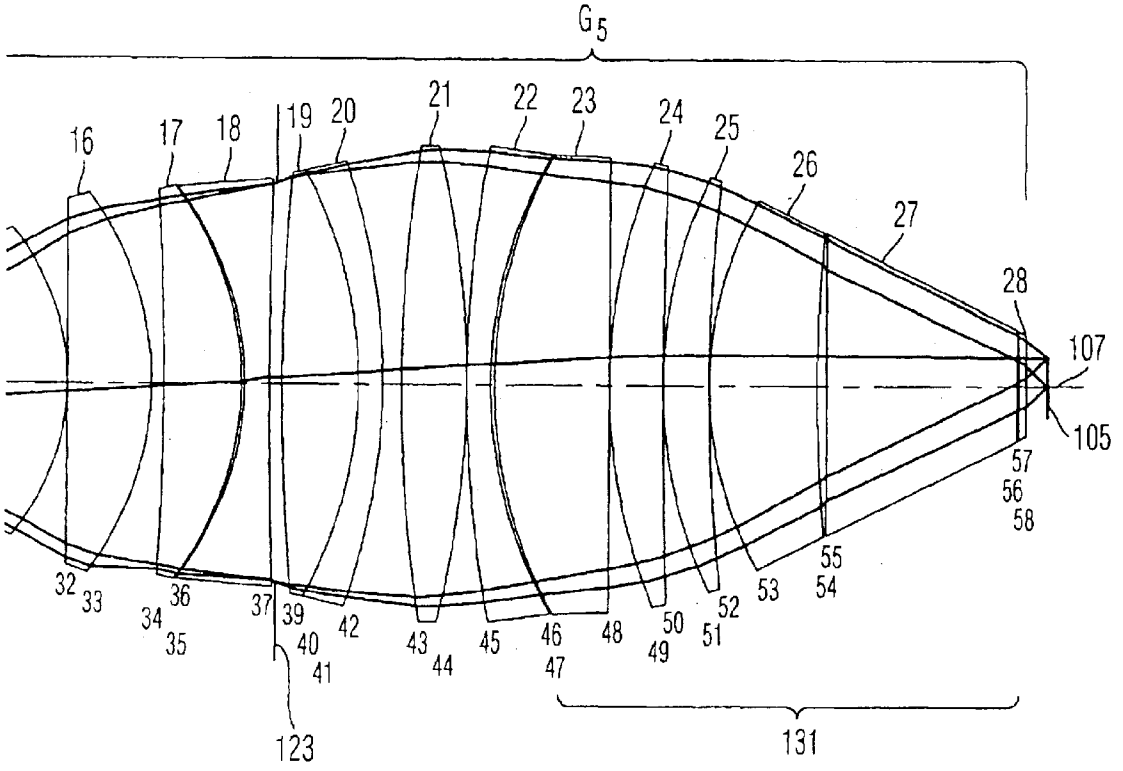

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Fig. 10b

PROJECTION EXPOSURE SYSTEM

The invention relates to a projection exposure system for imaging a first object in a region of a second object. In particular, the projection exposure system is provided to be used in a process, in particular, in a lithographic process for manufacturing miniaturized devices. Accordingly, the projection exposure system is configured to enable an imaging with comparatively high resolution and correspondingly high numerical aperture.

In a conventional step of a lithographic process, an image of the geometry of the structure to be produced is provided as the first object or a mask or reticle. The mask is illuminated with light of a first wavelength band having a width $\delta\lambda$ about a central working wavelength λ. With this light, the projection exposure system images the mask on the second object which is usually a wafer coated with a radiation-sensitive layer or resist. After having been exposed, the radiation-sensitive layer is subjected to further lithographic steps in order to form the miniaturized structures in the wafer in further process steps. The imaging of the mask on the radiation-sensitive layer is usually effected reduced in size, for example, with a scale of 4:1.

The present projection exposure system is particularly configured for effecting the illumination and imaging with light of a relatively wide wavelength band. Accordingly, the objective must have a relatively high chromatic correction in order to ensure also with such wide wavelength bands a highly resolved and sufficiently error-free imaging. A relative width $\delta\lambda/\lambda$ of the wavelength band particularly aimed at is larger than 0.005. As a result, the projection exposure system is particularly suited for light of the Hg-I line which comprises light of a spectrum of a wavelength band of 365.5 nm ±2 nm. However, the present projection exposure system is not limited to the use of the Hg-I line. Rather, the use of other radiation sources and thus other wavelength bands is contemplated as well.

FIG. **1** schematically shows a beam path through an earlier design of a projection exposure system or objective **101** for imaging a mask positionable in an object plane **103** at an image plane **105**. The conventional objective **101** is provided for imaging with the Hg-I line and a wavelength λ of 365.5 nm, the imaging being effected with a demagnification of 4 to 1 and a numerical aperture NA=0.65. A plurality of lenses is disposed along an optical axis **107**. In FIG. **1**, a continuous numbering of the lenses is indicated above the same proceeding from an object plane **103** to an image plane **105**. Below the lenses, there is indicated a continuous numbering of the refractive surfaces of the lenses in the same order. Here, the lens surface of the first lens disposed towards the object plane **103** is designated by 2. The surface of said lens disposed towards the image plane **105** is designated with 3 and the lens surface of lens **2** disposed towards the object plane **103** is designated by 4 etc.

In FIG. **1** there is further indicated: an upper marginal beam **109** as well as a lower marginal beam **111** of a field point **113** disposed on the optical axis **107** and an upper marginal beam **115**, a main beam **117** and a lower marginal beam **119** of an outermost field point **121**, respectively. Furthermore, a system diaphragm **123** is shown wherein the upper marginal beams **109** and **115** of the central field point **113** and the outermost field point **121**, respectively, approximately intersect. The lower marginal beams **111** and **119** of the field points **113** and **121**, respectively, approximately intersect therein as well.

As is evident from FIG. **1**, the depicted objective is a so-called "three-bulge system". This means that beam cross-sections of the imaging light comprise three expanded portions and, accordingly, two constricted portions disposed between the expanded portions. These constricted portions are referred to as beam waists, whereas the expanded portions are referred to as beam bulges. This beam path with three bulges and two waists is formed by accordingly arranging lens groups successively, wherein the lens groups in which the beam path is bulged predominantly provide a positive refractive power and, correspondingly, lens groups in which the beam path is constricted predominantly provide a negative refractive power. In FIG. **1**, these lens groups are referred to hereafter as G_1, G_2, G_3, G_4 and G_5, the lens groups G_1, G_3 and G_5 providing a positive refractive power and the lens groups G_2 and G_4 providing a negative refractive power.

The lenses indicated in hatched outline with lens numbers **1**, **3**, **9**, **13**, **14**, **15**, **16**, **22**, **27**, **30** and **31** are made of a high-dispersion material similar to a flint glass. The other lenses **2**, **4**, **5**, **6**, **7**, **8**, **10**, **11**, **12**, **17**, **18**, **19**, **21**, **23**, **24**, **25**, **26**, **28** and **29** are made of a low-dispersion material similar to a crown glass.

Although, according to calculations, the above-described earlier design of a projection exposure system exhibits satisfactory imaging properties as far as the numerical aperture and aberrations are concerned, comparative experiments performed on components of the system showed that the quality of the imaging through the components and the system, respectively, could diminish during operation in the course of time, in particular, when a high production rate is applied in the exposure of wafers, for example.

Accordingly, it is an object of the present invention to provide a projection exposure system of the above-described type which provides specific imaging properties with increased stability over time.

In this respect, the invention proceeds from a projection exposure system positionable between a first object and a second object for imaging the first object in a region of the second object with light of a wavelength band which has, in particular, a relative width $\delta\lambda/\lambda$ of more than 0.002, preferably, of more than 0.005. Such a width of the wavelength band is relatively large in so far as specific wavelength bands of radiation which is likewise conventionally applied in lithography and which is, for example, provided by lasers, are considerably narrower.

Furthermore, the projection exposure system is a so-called three-bulge system, that is, it comprises five lens groups, wherein, in the order of the arrangement of the lens groups between the first object and the second object, each one of a first, a third and a fifth lens group has, as a whole, a positive refractive power, and each one of the lens groups respectively disposed therebetween, that is, the second and the fourth lens group, each has, as a whole, a negative refractive power.

Due to the wide wavelength band to be used, the projection exposure system exhibits a chromatic correction. To this end, the materials used to manufacture the individual lenses are selected from two material groups. A first material group thereof comprises materials having Abbe numbers which are higher than a limit value and, accordingly, materials of a second material group have Abbe numbers which are lower than the limit value. This means, that materials of the first material group have a lower dispersion than materials of the second material group. At least one of the lenses of the projection exposure system is made of a material of the first material group and has a positive refractive power, whereas at least another one of the lenses of the projection exposure system is made of a material of the second material group and has a negative refractive power.

The invention is based on the finding that certain glasses suitable for use at the predetermined working wavelength have a reduced long-term stability than other ones of such glasses. Due to the light passing through the lens material during operation, changes occur in the lens materials which can also change the optical properties of the lenses. Known mechanisms of such a kind are mechanisms known as "compaction", "lens heating" or also "solarization". It has now been found that, in particular, those glasses exhibit a reduced long-term stability which have a relatively high refractive index and a high dispersion.

Based on this finding, the invention is distinguished, according to a first aspect, in that, within the fifth lens group, that is, the lens group disposed closest to the second object, there is provided a partial group of at least three directly successively arranged lenses which are made of a material included in the first material group. In particular, this partial group comprises those lenses of the fifth lens group which are disposed relatively close to the second object. In particular, said partial group also comprises that lens of the projection exposure system which is disposed as terminal lens closest to the second object. According to the invention, this does not conflict with the fact that certain further elements, such as a protective plate or the like, which are traversed by the beam path and have a comparatively small optical effect are disposed between such a terminal lens and the second object. This may be, for example, a plane-parallel plate or a lens having a low refractive power which has a thickness on the optical axis of less than 1% of a distance between the first object and the second object. Such a plane-parallel plate provides a certain spherical over-correction as optical effect and thus reduces certain image errors, such as the spherical aberration.

The lenses of the fifth lens group close to the second object have a small diameter as compared to other lenses of the system. Accordingly, the entire illumination intensity is passed through a relatively small cross-sectional area so that relatively high radiation densities are encountered in this region of the system. Accordingly, the lenses close to the second object are particularly susceptible to deteriorations of the optical properties thereof at long operation times, because it has been found that the change in the optical properties increases as the radiation density and the illumination time increases. According to the invention, these lenses are not made of materials of the second material group. The materials of the second material group, whose Abbe numbers are smaller than those of the first material group have mostly a higher refractive index than the materials of the first material group. At the same time, the materials of the second material group often exhibit a higher absorption than those of the first material group. Accordingly, the materials of the second material group are mostly more susceptible to changes caused by a high radiation load. Now, as the lenses of the projection exposure system which are close to the second object are made of materials of the first material group, the system exhibits a long-term stability which is increased as compared to the conventional system.

The partial lens group which is disposed close to the second object and made of materials of the first material group comprises as many lenses as possible, in particular, four lenses, preferably, five lenses and, still further preferred, at least six lenses.

Preferably, the portion of materials of the second material group is as low as possible in the system as a whole as well. Nevertheless, a certain portion of lenses which is made of materials of the second material group is necessary in order to provided a desired chromatic correction. It has been found to be favorable that the materials of the second material group used in the system account for less than 20 percent, preferably, less than 15 percent, referred to a volume portion of the processed materials of the first material group.

In light of the necessary chromatic correction, it is, however, not possible to make all lenses of the fifth lens group of materials of the first material group. It is thus necessary to provide, after the partial group made of materials of the first material group, at least one lens which is made of a material of the second material group. It has been found to be favorable for this lens made of the material of the second material group to have a free diameter which is as large as possible so that, due to the relatively large diameter, the radiation density is reduced at the location of this lens.

By free diameter of a lens, it is understood the diameter within the lens which is maximally available to a beam path traversing the objective. Accordingly, the free diameter of a lens basically corresponds to the overall diameter minus the region covered, for example, by the lens mount.

In regard to a maximum free diameter D_{max} of all lenses used in the system, the lens adjacent to the partial group and made of the material of the second material group has a free diameter which is larger than 0.7 times D_{max} and which is preferably larger than 0.8 times D_{max}.

Furthermore, it has been found to be favorable for lenses made of the second material group which are indispensable for chromatic correction to be disposed at locations in the beam path of the system where there is a radiation intensity which is distributed across the cross-sectional area as homogeneously as possible. Preferably, such locations are located in a region along the optical axis wherein bundle cross-sections of beams emanating from different field points of the first object are displaced from each other to a relatively little degree. It has been found to be favorable for this region to be selected such that the displacement of two beam bundles, relative to the free diameter, has a value of less than 0.1, one of said beam bundles emanating from a first outermost field point at the first object and a second one emanating from an outermost field point which is diametrically opposed to the first field point.

This region usually includes the location where a system diaphragm is disposed. Furthermore, it has been found to be advantageous for the region to be selected such that such a displacement between the two beam bundles of two diametrically opposed outermost field points, relative to the free diameter of the two beam bundles, is smaller than 0.05 at locations which are disposed between the system diaphragm and the second object.

By fabricating the lenses of the partial group disposed close to the second object of materials of the first material group which exhibit a relatively low susceptibility to radiation-induced changes, it is possible and preferably also provided for that at least one lens of this partial group is formed as a particularly thick lens because with such thick lenses it is also possible to correct specific aberrations. Accordingly, at least one lens of the partial group has preferably a maximum thickness along the optical axis which is larger than 0.3 times and preferably larger than 0.45 times the free diameter of this lens.

Furthermore, it has been found that lenses of the fourth lens group, too, that is, lenses of the second beam waist are sensitive to radiation-induced changes. On the other hand, the second beam waist is also a place where lenses having a negative refractive power and made of materials of the second material group are disposed, said lenses being necessary for the required chromatic correction.

According to a second aspect of the invention, it is provided for in a projection exposure system which comprises in its second beam waist lenses made of the second material group that said lenses are at least made of a material whose refractive index is smaller than 1.59 at a wavelength of 365.5 nm. Accordingly, at least such materials of the second material group are disposed in the second beam waist which, as compared to other materials of the second material group, have a relatively small refractive index and a lower radiation sensitivity.

As already explained above, particularly materials with a high refractive index have been found to be susceptible to radiation-induced changes. According to this aspect, the invention thus provides a projection exposure system which exhibits a relatively little susceptibility to radiation-induced changes when lenses made of materials of the second material group are disposed in the second beam waist.

It has been found to be extremely difficult to remove lenses made of materials of the second material group which are indispensable for the required chromatic correction from further regions of the projection exposure system which are different from the third bulge or fifth lens group.

According to a further aspect, the invention suggests providing lens groups which are different from the fifth lens group such that all lenses thereof are made of materials of the first material group and compensating for the thus accepted loss of chromatic correction involved therewith in that at least one lens of said lens groups has an aspherical surface. The fabrication of aspherical lens surfaces is much more complex and accordingly involves considerably higher costs than the fabrication of spherical lens surfaces. Nevertheless, in the present case, the invention offers a favorable compromise between fabrication complexity and long-term stability of the optical system. For it has been found that merely only very few lens surfaces must be aspherically in order to keep relatively large portions of the system free of lenses made of materials of the second material group. Preferably, two, in particularly, three, and more preferred, all of the lenses of the lens groups which are different from the fifth lens group comprise at most one lens with an aspherical surface. Moreover, at least one of the lens groups which are different from the fifth lens group, i.e., the first, the second, the third or/and the fourth lens group is then preferably completely free of lenses made of materials of the second material group.

In this respect, it is further preferred to provide the aspherical lens surface at such a location in the beam path where the aspherical surface has a maximum influence on the correction of aberrations. It has been found that such locations are positioned at boundaries between adjacent lens groups. Accordingly, a lens which is an outermost marginal lens on the optical axis of the respective lens group is preferably selected to have the aspherical surface. It is then further preferred to provide the aspherical surface on the side of this marginal lens which is oriented away from the lens group.

In order to define a distinction between lenses of two different lens groups, it is suggested to consider the path of an outer marginal beam of an outermost field point in the beam path. At the three bulges of the system, the path of said marginal beam is curved inwardly, i.e., towards the optical axis. Accordingly, it is curved outwardly, i.e., away from the optical axis at the two waists of the system. As a result, the path comprises a deflection point between the outwardly curved portions and the inwardly curved portions. With this approach it is in most cases possible to clearly allocate a lens in question to one of the lens groups. However, in some cases a first surface of a specific lens causes the marginal beam to be deflected away from the optical axis, while the other lens surface thereof already directs the beam towards the optical axis. Accordingly, the effect of the one surface could be allocated to the beam waist, whereas the effect of the other lens surface is allocated to the beam bulge. However, as a whole, such a lens is allocated to the beam waist or the second and fourth lens group, respectively, if the deflecting effect away from the optical axis of the one surface exceeds the deflecting effect towards the optical axis of the other lens surface and vice versa.

An alternative way of defining a distinction between lenses of two different lens groups relates to the cumulated refractive power of a lens group. Accordingly, a lens group is interpreted as a waist if it has, as a whole, a diverging effect and its overall refractive power is higher than the refractive power of that single lens of the system which has the highest diverging effect. Moreover, the amount of the overall refractive power of a waist is to be maximal, i.e., the negative overall refractive power of the lens group would increase again if lenses adjacent to the group were incorporated or a marginal lens of the lens group was omitted. In other words, this also means that the marginal lenses of a waist are each diverging lenses.

Bulges can be defined in analogous manner. They have a converging effect and have an overall refractive power which is higher than the refractive power of that single lens of the system which has the highest converging effect. Moreover, the overall refractive power of a bulge is to be maximal, i.e., the overall positive refractive power of the lens group would decrease again if lenses adjacent to the group were incorporated or a marginal lens of the lens group was omitted. In other words, this means that the marginal lenses of a bulge are each converging lenses.

As a whole, the projection exposure system comprises exactly three bulges and two waists. Only those three lens groups fulfilling the above-indicated criteria are interpreted as bulge which, among all lens groups fulfilling the above-indicated criteria, exhibit the highest positive overall refractive powers. The analogue applies to the two waists.

Preferably, all lenses of the first, second and third lens group are made of materials of the first material group. In this respect, it is even more preferred to make also all lenses of the fourth lens group of such materials.

However, good results are also obtained if the portion of lenses made of the second material group and disposed in lens groups in front of the second beam waist, that is, in the first, second or third lens group, account for a relatively small portion of the entire volume of lenses made of materials of the second material group. A favorable value is obtained if the lenses made of materials of the second material group and provided in the first, second and third lens group account for less than 10 percent, based on the entire volume of lenses made of materials of the second material group.

According to a further aspect of the invention, the fifth lens group, i.e., the lens group disposed closest to the second object, comprises a pair of lenses which in this application is referred to as pseudo-cemented lens pair. The pseudo-cemented lens pair consists of two directly adjacent lenses whose lens surfaces disposed towards each other are spaced apart from each other over their entire cross-section only by a very small distance. The distance between the two lens surfaces disposed towards each other is less than 2.50%, in particular, less than 1.25 of the free diameter of the largest one of the two lenses. One lens of the lens pair is made of a material of the first material group and has a positive

refractive power, while the other lens of the lens pair is made of a material of the second material group and has a negative refractive power. The lens surface of the lens of negative refractive power which is disposed towards the other lens is a concave surface.

Preferably, the other lens surface of the lens of negative refractive power disposed away from the other lens of the pair is a convex lens surface.

Moreover, it has been found to be advantageous for two of such pseudo-cemented lens pairs to be positioned between a system diaphragm of the optical system and the second object.

In this respect, it is equally favorable for at least one of said pseudo-cemented lens pairs disposed between a system diaphragm and the second object to comprise lens surfaces disposed towards each other which are curved such that the centers of curvature of the two lens surfaces disposed towards each other are positioned on the side of the second object. Moreover, it has likewise been found to be favorable for the lens surfaces disposed towards each other of a pseudo-cemented lens pair which is disposed between a system diaphragm and the first object to be curved such that the centers of curvature of the lens surfaces disposed towards each other are positioned on the side of the first object.

A favorable configuration of the system is obtained if a diaphragm is disposed in the region of the fifth lens group.

Moreover, the system is particularly suited for use of light of the Hg-I line.

The materials referred to herein as materials of the first material group preferably have a refractive index, measured at a wavelength of 365.5 nm, which is smaller than 1.55 and further preferred smaller than 1.53.

Accordingly, the glass of the company SCHOTT, Mainz, Germany, referred to as FK5 at the date of application is included in the first material group. Another definition of the first material group is thus possible in that all materials of the first material group have a refractive index which is smaller than 1.03 times that of the FK5 glass.

Vice versa, as materials of the second material group are selected such materials which have a refractive index at a wavelength of 365.5 nm which is higher than 1.54. Thus, in particular, the glass of the company SCHOTT referred to as LLF1 is included in the second material group. An alternative definition of the materials of the second material group is thus given in that all materials of the second material group are in a refractive index range which starts at 0.97 times the refractive index of the LLF1 glass and terminates at 1.07 times this refractive index.

However, the glass of SCHOTT referred to, at the date of application, as LF5 is, preferably, not contained in the second material group. Accordingly, an upper limit of the refractive index of the materials of the second material group can be defined by the refractive index of SCHOTT's LF5 glass. Accordingly, the materials of the second material group have a refractive index at the wavelength 365.5 nm which is smaller than 1.64 and which is, in particular, smaller than 1.59.

The limit value Vg which distinguishes materials of the first material group from that of the second material group as regards their Abbe numbers is preferably in the middle between the Abbe numbers of the FK5 and LF5 glasses of SCHOTT. Preferably, this limit value Vg is 400 if the Abbe number is determined at the wavelength λ=365.5 nm and a wavelength difference $\Delta\lambda$ of 3 nm.

Furthermore, the first material group preferably comprises quartz glass.

With a view to high-resolution applications, the numerical aperture of the system on the side of the second object is larger than 0.60, preferably, larger than 0.65 and, most preferred, larger than 0.70.

The quality of the optical imaging can likewise be characterized by the so-called Strehl number. For the present projection exposure system, the Strehl number is preferably higher than 0.95 and further preferred higher than 0.98.

Furthermore, such a high-resolution system is distinguished in that, at a given distance between the first object and the second object, there are provided relatively many lenses in order to keep the optical loads which the individual lens surfaces are subjected to, i.e., the beam deflections produced by the individual lens surfaces, as low as possible. Preferably, shortest distances of adjacent lens surfaces, that is, distances between lenses measured, depending on the lens type, on the optical axis or the lens edges are therefore smaller than 25 mm and further preferred less than 20 mm. Shortest distances between opposite lens surfaces, averaged over the projection exposure system, are smaller than 3 mm, preferably smaller than 1.5 mm.

In order to render the projection exposure system largely insensitive to slight misalignments between the first and the second object, the projection exposure system is preferably provided telecentrically on both sides.

Embodiments of the invention are described hereinafter with reference to Figures, wherein

FIG. **1** shows an earlier design of a projection exposure system,

FIG. **2** shows a first embodiment of a projection exposure system according to the invention,

FIG. **3** shows a cross-section through a beam path of a projection exposure system for illustrating a displacement of beam bundles,

FIG. **4** shows a second embodiment of a projection exposure system according to the invention,

FIG. **5** shows a third embodiment of a projection exposure system according to the invention,

FIG. **6** shows a fourth embodiment of a projection exposure system according to the invention,

FIG. **7** shows a fifth embodiment of a projection exposure system according to the invention,

FIG. **8** shows a sixth embodiment of a projection exposure system according to the invention,

FIG. **9** shows a seventh embodiment of a projection exposure system according to the invention, and

FIG. **10** shows an eighth embodiment of a projection exposure system according to the invention.

An embodiment of the projection exposure system according to the invention is illustrated in FIG. **2**.

Optical data of the projection exposure system shown in FIG. **2** are indicated below in Table 1, wherein the first column indicates the numbering of the respective refracting surfaces, the second column indicates the radii of curvature of the refracting surfaces, the third column indicates the axial distances between adjacent refracting surfaces, the fourth column indicates the material which is disposed between the refracting surfaces, the fifth column indicates the refractive index of this optical material measured at a wavelength of 365.5 nm and the sixth column indicates half of the free diameter of the lens allocated e respective refracting surface. The glasses used for the objective **101** are obtained by the company SCHOTT and, accordingly, SCHOTT's respective product names (state: autumn 2001) are indicated in the fourth column as glass type:

TABLE 1

Surface	Radii	Thicknesses	Medium	Refractive index	½ free diameter
0	0.000000000	32.000000000		1.00000000	59.000
1	0.000000000	3.410700000		1.00000000	64.523
2	−407.431645000	24.124404000	LF5	1.61906768	64.233
3	−190.763960000	9.051883000	HE20	0.99976000	67.088
4	−126.896809000	10.390974000	FK5	1.50393369	67.174
5	315.746411000	10.170796000	HE20	0.99976000	76.060
6	2642.002430000	26.521311000	LF5	1.61906768	77.158
7	−170.350740000	0.600000000	HE20	0.99976000	78.941
8	688.779159000	15.251492000	LLF1	1.57916392	83.139
9	−828.961373000	0.600000000	HE20	0.99976000	83.495
10	253.049406000	34.466538000	FK5	1.50393369	84.172
11	−247.629502000	0.500000000	HE20	0.99976000	83.655
12	134.142191000	14.197210000	FK5	1.50393369	70.861
13	95.532190000	27.880102000	HE20	0.99976000	63.005
14	−738.224398000	8.000000000	FK5	1.50393369	62.611
15	139.763218000	19.067021000	HE20	0.99976000	59.263
16	−825.440653000	9.000000000	FK5	1.50393369	59.338
17	189.026000000	36.028267000	HE20	0.99976000	60.125
18	−87.960623000	11.529505000	LLF1	1.57916392	60.608
19	−6356.790846000	0.612768000	HE20	0.99976000	77.138
20	1948.324738000	44.976940000	FK5	1.50393369	79.527
21	−112.462247000	0.705368000	HE20	0.99976000	82.798
22	−882.161402000	27.572454000	FK5	1.50393369	89.995
23	−233.533951000	0.600000000	HE20	0.99976000	92.374
24	474.300365000	22.466843000	FK5	1.50393369	94.138
25	−575.153971000	0.600000000	HE20	0.99976000	93.991
26	139.676984000	32.087368000	FK5	1.50393369	88.856
27	521.190488000	4.133506000	HE20	0.99976000	86.351
28	153.134948000	40.740797000	FK5	1.50393369	78.332
29	87.932647000	33.195673000	HE20	0.99976000	58.278
30	−186.852044000	8.000000000	LLF1	1.57916392	57.642
31	131.053345000	34.321763000	HE20	0.99976000	55.746
32	−94.087869000	17.769438000	LLF1	1.57916392	56.315
33	748.500214000	11.607438000	HE20	0.99976000	71.079
34	−434.234909000	22.494741000	FK5	1.50393369	72.711
35	−154.683362000	0.600000000	HE20	0.99976000	77.013
36	537.928136000	31.248045000	FK5	1.50393369	91.846
37	−355.890330000	1.284344000	HE20	0.99976000	94.426
38	2563.090901000	36.287956000	FK5	1.50393369	98.052
39	−199.037047000	1.700000000	HE20	0.99976000	99.134
40	0.000000000	6.000000000	SUPRA1	1.47447761	96.837
41	0.000000000	2.100000000	HE20	0.99976000	96.506
42	0.000000000	−2.100000000	HE20	0.99976000	96.356
43	0.000000000	9.000000000	HE20	0.99976000	96.506
44	314.514859000	31.425407000	FK5	1.50393369	98.490
45	−506.854040000	15.763950000	HE20	0.99976000	98.200
46	−216.541413000	5.000000000	LF5	1.61906768	97.844
47	367.750084000	3.498240000	HE20	0.99976000	101.637
48	389.909912000	38.818137000	FK5	1.50393369	102.687
49	−402.292074000	19.425707000	HE20	0.99976000	104.102
50	551.171957000	33.360959000	FK5	1.50393369	106.437
51	−340.874035000	0.600000000	HE20	0.99976000	106.254
52	132.912140000	37.466243000	FK5	1.50393369	93.647
53	540.348303000	0.600000000	HE20	0.99976000	90.723
54	119.075653000	24.827993000	FK5	1.50393369	77.436
55	232.598798000	11.889365000	HE20	0.99976000	72.129
56	1290.376947000	12.237054000	LLF1	1.57916392	69.928
57	146.400197000	0.600000000	HE20	0.99976000	58.321
58	73.325602000	14.970586000	FK5	1.50393369	52.371
59	53.686391000	13.134189000	HE20	0.99976000	42.587
60	98.428877000	14.235259000	FK5	1.50393369	41.873
61	−1373.070799000	3.329224000	HE20	0.99976000	39.821
62	−380.732215000	5.841963000	FK5	1.50393369	38.039
63	52.895408000	0.600000000	HE20	0.99976000	30.452
64	43.341616000	16.091676000	FK5	1.50393369	29.613
65	1336.072484000	1.488402000	HE20	0.99976000	27.286
66	0.000000000	2.000000000	SUPRA1	1.47447761	26.229
67	0.000000000	12.000100000		1.00000000	25.232
68	0.000000000	0.000000000		1.00000000	14.752

The refractive indices of the media, namely gases and optical materials, relate to a pressure of 950 mbar at 22° C. They are normalized in respect of the refractive index n=1 of ambient air at a pressure of 1013.3 mbar at 20° C. Therefore, due to the lower pressure and the higher temperature, it is also possible that refractive indices of n<1 result. Corresponding numerical values are inferable from Table 2.

TABLE 2

Refractive indices of the gases and optical materials of the objective calculation at different wavelength at 950 mbar and 22° Celsius as compared to ambient air at 1013.3 mbar and 20° Celsius (n = 1)

	L710	LF5HL	HE20	FK5HL	LLF1HL	SIO2HL
368.15 nm	0.999982	1.61802054	0.99976	1.50351719	1.57824125	1.47410903
367.20 nm	0.999982	1.61837858	0.99976	1.50365701	1.57852927	1.47424480
366.25 nm	0.999982	1.61874045	0.99976	1.50379801	1.57882024	1.47438171
365.50 nm	0.999982	1.61902889	0.99976	1.50391017	1.57905206	1.47449062
364.75 nm	0.999982	1.61931978	0.99976	1.50402309	1.57928578	1.47460025
363.90 nm	0.999982	1.61965247	0.99976	1.50415198	1.57955298	1.47472539
362.80 nm	0.999982	1.62008782	0.99976	1.50432024	1.57990246	1.47488874

Along an optical axis **107**, there are successively disposed 31 lenses, each having spherical lens surfaces, followed by a plane-parallel terminal plate made of quartz glass with a thickness of 2 mm which is designated by reference number **32** in FIG. **2**. The system serves to image a substantially planar object area **103** in an image region **105** which is likewise substantially planar. FIG. **2** furthermore shows an upper marginal beam **109** and a lower marginal beam **111** of a beam bundle emanating from a central field point **113** as well as a upper marginal beam **115**, a main beam **117** and a lower marginal beam **119** of an outermost lower field point **121**. It is evident from the path of the marginal beams **109**, **111**, **115** and **117** that the projection exposure system **101** is a three-bulge system. The 31 lenses or, with the terminal plate **32** being incorporated, 32 optical elements of the system can be divided into five groups. Lenses **1** to **5** define a first lens group G_1 which, as a whole, provides a positive refractive power. Lenses **6** to **9** define a lens group G_2 which, as a whole, provides a negative refractive power. Lenses **10** to **13** define a third lens group G_3 which, as a whole, provides a positive refractive power. Lenses **14** to **16** define a fourth lens group G_4 which, as a whole, provides a negative refractive power. Lenses **17** to **31**, plus **32**, define a fifth lens group G_5which, as a whole, provides a positive refractive power. A system diaphragm **123** is disposed in the region of the fifth lens group G_5 where the lower marginal beam **111** of the central field point **113** intersects the lower marginal beam **119** of the lower field point **121**.

The following materials are used for the optical elements: the lenses **2**, **5**, **6**, **7**, **8**, **10**, **11**, **12**, **13**, **14**, **17**, **18**, **19**, **21**, **23**, **24**, **25**, **26**, **28**, **29**, **30** and **31** are made of the FK5 material of the company SCHOTT. The lenses **1**, **3** and **22** are made of the LF5 material of SCHOTT. The lenses **4**, **9**, **15**, **16** and **27** are made of the LLF1 material of SCHOTT. The components **20** and **30** are made of quartz glass.

The LF5 and LLF1 materials are allocatable to the group of flint glasses. The FK5 material is allocatable to the group of crown glasses.

The lenses **28**, **29** **30** and **31** which are close to the image plane **105**, that is, in operation, close to the wafer are made of the FK5 material. This material has a refractive index of 1.50393369 and thus of less than 1.55 if this refractive index is determined at a wavelength of 365.5 nm. Moreover, this material has an Abbe number $V_{365.5}$≈560 if it is determined according to the formula

$$V_{365.5} = \frac{(N_{365.5} - 1)}{N_{362.5} - N_{368.5}}$$

wherein N_x is the refractive index of the material at a wavelength x. The Abbe number of the FK5 material is thus higher than a limit value V_g of, for example, 400.

The FK5 material of SCHOTT belongs to the group of materials referred to in this application as first material group.

The LF5 material has a refractive index of 1.61906768 and an Abbe number of about $V_{365.5}$≈270 if it is likewise determined according to the above formula.

The LLF1 material has a refractive index if 1.57916392 and an Abbe number of $V_{365.5}$≈310 if it is likewise determined according to the above formula.

The refractive indices of the LF5 and LLF1 glasses are thus higher than 1.54, while the Abbe numbers thereof are smaller than the exemplary limit value of V_g=400.

Of the materials used, the FK5 material exhibits the second best long-term stability against radiation of the wavelength band about 365.5 nm for which the system is predominantly designed. The best stability is exhibited by the components **20** and **32** made of quartz glass.

The material LLF1 exhibits a stability against radiation load which is reduced as compared to FK5, whereas the LF5 material exhibits the lowest long-term stability of all materials. Of the components **1**, **3**, **4**, **9**, **15**, **16**, **22** and **27** made of the second material group, merely lens **22** is made of the LF5 material. This lens, however, is one of the lenses of the system with the largest free diameter, so that, due to the large cross-sectional area, a reduced radiation load occurs. Lens **22** made of the LF5 material with a free diameter of 195.69 mm has a size of 91.9% of the lens **24** which, having a free diameter of 212.87 mm, is the lens having the largest free diameter in the system.

Moreover, lens **22** made of the LF5 material is disposed at a location of the beam path where there is a relatively little displacement of bundle cross-sections of beam bundles emanating from diametrically opposed field points.

This is illustrated in further detail in FIG. **3**. In this Figure, D indicates half of the free diameter of lens **22**, r_1 indicates the radius of the beam bundle which emanates from the lower outermost field point **121** and r_2 indicates the radius of a beam bundle which emanates from a further outermost field point **121** disposed diametrically opposed to the lower outermost field point **121** in respect of the optical axis **107**.

A distance between bundle centers **131** and **132** of these two beam bundles is designated by q in FIG. **3**. In FIG. **3**, a ratio of q to 2D is about 20° which does not necessarily corresponds to the real conditions, for example, in lens **22**.

In the following, variants of the embodiment described with reference to FIGS. **2** and **3** will be described. Components which correspond in structure and function to components of the embodiment described with reference to FIGS. **2** and **3** are designated by the corresponding reference signs. FIG. **4** illustrates a further projection exposure system **101**, the optical data of which are indicated in the following Table 3:

TABLE 3

Surface	Radii	Thicknesses	Medium	Refractive index	½ free diameter
0	0.000000000	32.000000000	L710	0.99998200	59.000
1	0.000000000	5.201908388	L710	0.99998200	64.613
2	−661.054742744	10.597577126	LLF1HL	1.57905206	64.964
3	−244.763741563	10.881589768	HE20	0.99976000	65.673
4	−130.229653149	8.000000000	FK5HL	1.50391017	65.818
5	359.452379850	8.145832344	HE20	0.99976000	73.729
6	1869.000000000	25.889017376	LF5HL	1.61902889	74.786
7	−169.172054615	0.600000000	HE20	0.99976000	76.510
8	775.360747172	15.383724400	LLF1HL	1.57905206	80.033
9	−494.373044356	0.600000000	HE20	0.99976000	80.300
10	216.147437120	32.669070619	FK5HL	1.50391017	80.192
11	−302.103981085	0.600000000	HE20	0.99976000	79.296
12	192.039294254	14.309261863	FK5HL	1.50391017	70.109
13	112.442556683	20.570062640	HE20	0.99976000	62.479
14	4969.718406730	5.000000000	FK5HL	1.50391017	61.975
15	130.169698546	22.263540496	HE20	0.99976000	58.818
16	−298.524605304	7.171571120	FK5HL	1.50391017	58.803
17	139.540198572	0.760163000	HE20	0.99976000	60.492
18	137.246786622	11.000000000	SIO2HL	1.47449062	60.996
19	177.468981757	41.134978753	HE20	0.99976000	61.523
20	−86.795812248	6.783726889	LLF1HL	1.57905206	62.581
21	−1084.641004990	1.225007423	HE20	0.99976000	78.922
22	−2819.674356910	47.217471701	FK5HL	1.50391017	81.744
23	−113.563780062	0.600000112	HE20	0.99976000	86.364
24	0.000000000	28.724827557	FK5HL	1.50391017	98.367
25	−226.420099890	0.600000169	HE20	0.99976000	99.467
26	449.900971158	23.762201304	FK5HL	1.50391017	101.252
27	−731.539953780	0.600000000	HE20	0.99976000	101.024
28	206.572700755	22.356910583	FK5HL	1.50391017	96.647
29	715.807052636	0.812729044	HE20	0.99976000	95.399
30	115.626477487	43.663887059	FK5HL	1.50391017	84.921
31	96.099626638	10.611297966	HE20	0.99976000	66.739
32	127.905996862	10.000000000	FK5HL	1.50391017	65.999
33	88.119735529	37.471785738	HE20	0.99976000	59.453
34	−161.423801442	6.000000000	LLF1HL	1.57905206	58.614
35	156.517030870	30.334748725	HE20	0.99976000	58.544
36	−118.649853783	6.642688287	LLF1HL	1.57905206	59.312
37	466.706723085	12.854093749	HE20	0.99976000	69.368
38	−463.523475033	16.570891307	FK5HL	1.50391017	71.208
39	−174.413396988	0.649146799	HE20	0.99976000	74.263
40	614.751431159	28.470734271	FK5HL	1.50391017	88.573
41	−267.716429052	8.900070186	HE20	0.99976000	90.466
42	513.314914536	38.676154625	FK5HL	1.50391017	99.799
43	−281.490221229	2.000000000	HE20	0.99976000	100.864
44	0.000000000	6.000000000	SIO2HL	1.47449062	100.153
45	0.000000000	16.000000000	HE20	0.99976000	100.013
46	0.000000000	−16.000000000	HE20	0.99976000	99.464
47	0.000000000	14.000000000	HE20	0.99976000	100.013
48	501.198000000	32.719926806	FK5HL	1.50391017	100.430
49	−316.233100115	3.222987363	HE20	0.99976000	100.483
50	−293.599499791	9.700000000	LF5HL	1.61902889	100.158
51	219.350000000	14.313216090	HE20	0.99976000	102.388
52	278.199999792	50.848784677	FK5HL	1.50391017	107.794
53	−293.599499920	0.600000000	HE20	0.99976000	108.890
54	321.720000155	31.720910762	FK5HL	1.50391017	109.046
55	−850.070000069	0.600000000	HE20	0.99976000	108.376
56	128.878000341	39.754795339	FK5HL	1.50391017	95.863
57	481.232000069	0.600000000	HE20	0.99976000	92.881
58	119.616000000	20.200550207	FK5HL	1.50391017	78.628
59	184.122999990	14.243725202	HE20	0.99976000	74.005
60	793.959999987	12.591473480	LLF1HL	1.57905206	72.131
61	142.158000423	0.600000000	HE20	0.99976000	60.052
62	97.579000000	18.237940745	FK5HL	1.50391017	56.849
63	58.110000000	11.497956939	HE20	0.99976000	44.206
64	91.829000000	24.368245501	SIO2HL	1.47449062	43.385
65	56.848000000	0.600000000	HE20	0.99976000	32.677
66	45.041000000	17.912233519	SIO2HL	1.47449062	31.580
67	1150.000000000	2.360581982	HE20	0.99976000	28.796
68	0.000000000	3.000000000	SIO2HL	1.47449062	27.006
69	0.000000000	12.000000000	L710	0.99998200	25.485
70	0.000000000				14.750

All of the 32 lenses of the projection exposure system shown in FIG. **4** have spherical surfaces. Lenses **1** to **5** are allocated to a first lens group G_1 having a positive refractive power. Lenses **6** to **10** are allocated to a second lens group G_2 having a negative refractive power. Lenses **11** to **14** are allocated to a third lens group G_3 having, as a whole, a positive refractive power. Lenses **15** to **18** are allocated to a fourth lens group G_4 having, as a whole, a negative refractive power. Lenses **19** to **32**, plus a terminal plate **33**, are allocated to a fifth lens group G_5 having, as a whole, a positive refractive power.

The system shown in FIG. **4** is very similar to the system in FIG. **2**. However, it differs from the latter in of a partial group **131** of the fifth lens group G_5 which is close to the second object **105** and comprises the lenses **30**, **31** and **32**, two lenses, namely lenses **31** and **32**, are made of quartz glass. However, none of the lenses **30**, **31**, **32** of the lens group **131** disposed close to the wafer is made of materials of the second material group. In the embodiment shown in FIG. **4**, too, relatively few components are made of flint glass, here glasses LLF1 and LF5. Lens **24** which is made of the LF5 material, i.e., the material of the two above-mentioned ones which is more susceptible to radiation load, is disposed in the region of a relatively large free beam diameter of the fifth lens group G_5.

A further embodiment of the projection exposure system is shown in FIG. **5**, the optical data thereof being indicated in Table 4 below:

TABLE 4

Surface	Radii	Thicknesses	Medium	Refractive index	½ free diameter
0	0.000000000	32.000000000		1.00000000	59.000
1	0.000000000	20.661283629		1.00000000	64.558
2	−111.188782764	7.500000000	SUPRA1	1.47447761	64.558
3	1044.117026287	5.631557530		1.00000000	74.836
4	−6522.757198827	28.462316410	LF5	1.61906768	76.297
5	−152.064070605	0.700000000		1.00000000	78.632
6	4174.666996920	18.820233303	LF5	1.61906768	82.895
7	−323.399070672	0.700000000		1.00000000	83.618
8	276.085518412	28.324664425	FK5	1.50393369	83.695
9	−417.240395857	0.700000000		1.00000000	82.972
10	197.537123613	18.665436059	SUPRA1	1.47447761	76.341
11	747.689679031	0.700000000		1.00000000	74.115
12	117.405251606	8.370816389	LF5	1.61906768	66.458
13	88.004919252	25.692310808		1.00000000	60.325
14	2407.809906857	7.500000000	FK5	1.50393369	59.536
15	112.869324513	23.413285790		1.00000000	55.037
16	−232.219785599	7.500000000	LLF1	1.57916392	54.911
17	200.734425625	29.730361848		1.00000000	56.181
18	−93.822277301	34.992229843	LF5	1.61906768	56.880
19	16329.783148621	0.700000000		1.00000000	81.243
20	1496.565112315	41.754230352	FK5	1.50393369	83.779
21	−133.013546131	0.700000000		1.00000000	87.012
22	−997.788220363	25.716111377	FK5	1.50393369	95.340
23	−213.328508971	0.700000000		1.00000000	96.909
24	565.024020217	22.267630949	LLF1	1.57916392	99.826
25	−783.111728915	0.700000000		1.00000000	99.755
26	156.357415717	40.532446738	FK5	1.50393369	96.086
27	−8404.755315777	0.700000000		1.00000000	93.963
28	162.727735224	28.087985964	FK5	1.50393369	82.323
29	2433.769683095	7.033622247		1.00000000	78.656
30	−894.969351597	9.030606766	FK5	1.50393369	76.346
31	91.319408938	33.822770472		1.00000000	60.343
32	−212.432282527	7.500000000	FK5	1.50393369	59.647
33	124.212404197	31.911524352		1.00000000	57.418
34	−116.574231997	27.624156767	LF5	1.61906768	57.812
35	569.538912765	9.728886180		1.00000000	72.438
36	−1057.564068923	22.838848295	FK5	1.50393369	74.166
37	−171.030233916	0.700000000		1.00000000	77.256
38	366.961198433	28.192949883	FK5	1.50393369	89.665
39	−414.260486992	0.700000000		1.00000000	90.788
40	384.366160426	25.455343714	FK5	1.50393369	93.663
41	−641.891335386	−2.766111080		1.00000000	93.658
42	0.000000000	5.636846798		1.00000000	93.346
43	431.031845480	37.977337508	FK5	1.50393369	94.668
44	−228.618310317	3.199548355		1.00000000	94.516
45	−218.508619872	23.845728106	LF5	1.61906768	93.789
46	240.366071551	13.230649262		1.00000000	95.133
47	522.257860819	31.407780366	FK5	1.50393369	95.987
48	−311.265656291	0.700000000		1.00000000	97.033
49	246.201672686	35.112104162	FK5	1.50393369	99.331
50	6270.516480178	0.700000000		1.00000000	97.755
51	155.310012192	46.384312484	FK5	1.50393369	93.073
52	477.890701308	0.700000000		1.00000000	85.347
53	176.712993621	27.621391260	FK5	1.50393369	80.215
54	266.563889614	0.700000000		1.00000000	72.023
55	128.812275863	58.534298914	FK5	1.50393369	66.871
56	387.199528908	7.439347123		1.00000000	43.077

TABLE 4-continued

Surface	Radii	Thicknesses	Medium	Refractive index	½ free diameter
57	−329.427177242	7.500000000	FK5	1.50393369	40.794
58	53.414291554	0.700000000		1.00000000	31.925
59	47.258242704	19.426754985	FK5	1.50393369	31.342
60	−732.790228385	1.488401667		1.00000000	28.072
61	0.000000000	2.000000000	SUPRA1	1.47447761	26.292
62	0.000000000	12.001007801		1.00000000	25.291
63	0.000000000	−0.001150594		1.00000000	14.750

An optical system **101** shown in FIG. **5** comprises 29 lenses, each having spherical lens surfaces, as well as a terminal plate **30**. The classification of these lenses into lens groups having a positive refractive power and a negative refractive power is as follows:

Lens **1** to **5** are allocated to a first lens group G_1 having a positive refractive power. Lenses **6** to **9** are allocated to a second lens group G_2 having a negative refractive power. Lenses **10** to **14** are allocated to a third lens group G_3 having, as a whole, a positive refractive power. Lenses **15** to **17** are allocated to a fourth lens group G_4 having, as a whole, a negative refractive power. Lenses **18** to **29**, plus a terminal plate **30**, are allocated to a fifth lens group G_5 having, as a whole, a positive refractive power.

The fifth lens group G_5 merely comprises a single lens (number **22**) which is made of a material of the second material group. This is the LF5 material. This lens is again disposed in a region where a displacement of the bundle cross-sections of beams which emanate from diametrically opposed outermost field points is very small and has a value of less than 10% of the free diameter of this lens. A partial group **131** consisting of lenses **23** to **29** of lenses of the fifth lens group G_5 is made of the FK5 material. These are the lenses which are close to the second object or wafer and are subjected to an extremely high load both as regards radiation density and radiation deflection. These lenses are thus made of a material of the first material group which is particularly radiation-resistant.

An embodiment of a projection exposure system **101** shown in FIG. **6** comprises 32 optical components (the optical data thereof are indicated in Table 5 below). Their classification in groups G_1 to G_5 can be effected as follows:

Lenses **1** to **5** are allocated to a first lens group G_1 having a positive refractive power. Lenses **6** to **10** are allocated to a second lens group G_2 having a negative refractive power. Lenses **11** to **15** are allocated to a third lens group G_3 having, as a whole, a positive refractive power. Lenses **16** to **18** are allocated to a fourth lens group G_4 having, as a whole, a negative refractive power. Lenses **19** to **31**, plus lens **32**, are allocated to a fifth lens group G_5 having, as a a positive refractive power.

TABLE 5

Surface	Radii	Thicknesses	Medium	Refractive index	½ free diameter
0	0.000000000	32.000000000	L710	0.99998200	59.000
1	0.000000000	9.444238199	L710	0.99998200	64.535
2	−311.927851990	11.089934901	FK5	1.50393369	64.984
3	−181.428152021	10.400892620	HE20	0.99976000	66.009
4	−118.524477504	8.000000000	FK5	1.50393369	66.236
5	401.250423152	9.147444865	HE20	0.99976000	75.864
6	4344.106783837	27.681393443	LF5	1.61906768	77.396
7	−156.541320233	0.700000000	HE20	0.99976000	79.163
8	389.649615896	20.306038892	FK5	1.50393369	84.577
9	−654.208379945	0.700000000	HE20	0.99976000	84.776
10	287.218885965	31.151432162	FK5	1.50393369	84.559
11	−279.199609529	0.700000000	HE20	0.99976000	83.956
12	158.523597789	20.996036629	FK5	1.50393369	72.692
13	103.787825100	26.116106158	HE20	0.99976000	62.671
14	−782.918785418	7.000000000	FK5	1.50393369	61.997
15	131.910015514	21.785618927	HE20	0.99976000	58.731
16	−323.683912563	7.000000000	FK5	1.50393369	58.821
17	162.083641806	0.700000000	HE20	0.99976000	60.874
18	154.584107186	7.704332741	FK5	1.50393369	61.477
19	186.827774796	40.187765501	HE20	0.99976000	61.882
20	−86.225264321	7.000000000	LLF1	1.57916392	62.707
21	−705.417764820	4.794429538	HE20	0.99976000	78.952
22	−1040.227609005	43.042787088	FK5	1.50393369	84.413
23	−113.882639114	0.700000000	HE20	0.99976000	87.484
24	−1501.300440742	24.953541861	FK5	1.50393369	99.619
25	−234.915280384	0.700000000	HE20	0.99976000	100.878
26	510.926182463	26.022851665	FK5	1.50393369	104.709
27	−571.417608701	0.700000000	HE20	0.99976000	104.724
28	218.131483276	26.671165595	FK5	1.50393369	101.497
29	1296.198010698	0.700000000	HE20	0.99976000	100.230
30	120.687132319	56.529600371	FK5	1.50393369	88.992
31	90.620870733	11.750561859	HE20	0.99976000	65.101

TABLE 5-continued

Surface	Radii	Thicknesses	Medium	Refractive index	½ free diameter
32	121.167835192	7.000000000	FK5	1.50393369	64.239
33	90.915840804	35.617083391	HE20	0.99976000	59.635
34	−167.750316271	7.000000000	LLF1	1.57916392	59.022
35	150.866221845	31.250738202	HE20	0.99976000	58.993
36	−115.391399204	7.000000000	LLF1	1.57916392	59.691
37	532.903096962	9.549410572	HE20	0.99976000	70.580
38	−1179.176098456	22.240519284	FK5	1.50393369	72.887
39	−165.888015525	0.700000000	HE20	0.99976000	76.057
40	397.885948905	32.827416184	FK5	1.50393369	91.934
41	−270.366549631	5.911356416	HE20	0.99976000	93.170
42	618.783704027	26.820604247	FK5	1.50393369	97.881
43	−362.711535921	1.700000000	HE20	0.99976000	98.142
44	0.000000000	6.000000000	SUPRA1	1.47447761	97.065
45	0.000000000	16.000000000	HE20	0.99976000	96.777
46	0.000000000	−16.000000000	HE20	0.99976000	95.647
47	0.000000000	9.000000000	HE20	0.99976000	96.777
48	336.329780290	37.325480941	FK5	1.50393369	96.261
49	−284.881632662	7.861225022	HE20	0.99976000	95.975
50	−244.589205122	10.000000000	LF5	1.61906768	94.506
51	193.491898200	11.805473540	HE20	0.99976000	95.053
52	235.940524490	46.343334587	FK5	1.50393369	99.014
53	−271.608864830	0.700000000	HE20	0.99976000	99.562
54	383.703940087	21.747494278	FK5	1.50393369	97.901
55	−1334.216744204	0.700000000	HE20	0.99976000	97.232
56	119.914859771	35.236397604	FK5	1.50393369	87.691
57	430.327205454	0.700000000	HE20	0.99976000	85.077
58	112.713649998	22.770868262	FK5	1.50393369	73.087
59	191.347588922	12.870171550	HE20	0.99976000	67.414
60	1059.409420929	30.749249456	FK5	1.50393369	65.040
61	55.209590077	8.161805322	HE20	0.99976000	40.284
62	71.322184756	17.306203742	SUPRA1	1.47447761	39.436
63	57.924369902	0.700000000	HE20	0.99976000	32.783
64	45.290125679	18.322003097	SUPRA1	1.47447761	31.685
65	3911.657028771	2.706991290	HE20	0.99976000	28.931
66	0.000000000	3.000000000	SUPRA1	1.47447761	26.670
67	0.000000000	12.000000000	L710	0.99998200	25.180
68	0.000000000	0.000000000			14.750

A partial group **131** consisting of lenses **25** to **32** and being disposed closest to the object or wafer is made of the FK5 material. Merely one lens of the lens group G_5, namely lens **24**, is made of a material of the second material group, y the LF5 material. This is again a lens having a very free diameter.

A third lens group G_3 is completely free of materials of the second material group.

The first bulge, that is, the first lens group G_1 and the first waist, that is, the second lens group G_2, merely comprise a single lens which is made of a material of the Merely the second material group, namely LF5 and LLF1, respectively.

Merely the second waist corresponding to the fourth lens group comprises two lenses which are made of the LLF1 material of the second material group. Accordingly, the system **101** includes a particularly small volume proportion of materials of the second material group.

Moreover, the lenses **30** and **31** which are disposed closest wafer are made of quartz glass.

A further embodiment of a projection exposure system **101** is shown in FIG. **7**, the optical data of which are indicated in Table 6 below:

TABLE 6

Surface	Radii	Thicknesses	Medium	Refractive index	½ free diameter
0	0.000000000	32.000000000	L710	0.99998200	54.410
1	0.000000000	6.509726858	L710	0.99998200	59.909
2	−345.837608233	20.937634434	FK5	1.50393369	60.123
3	−138.710240702	4.139495305	HE22	0.99975600	62.120
4	−117.590857172	7.000000000	FK5	1.50393369	62.137
5	−231.054862836	0.700000000	HE22	0.99975600	65.357
6	−579.679241805	28.686881101	FK5	1.50393369	66.421
7	−137.166922545	0.700000000	HE22	0.99975600	68.551
8	120.364915645	24.197329481	FK5	1.50393369	66.273
9	1050.735539941	0.700000000	HE22	0.99975600	64.707
10	199.473249572	7.000000000	FK5	1.50393369	61.726
11	100.241943039	11.872055781	HE22	0.99975600	56.673
12	220.743162395	7.000000000	FK5	1.50393369	56.366
13	109.954107613	20.291682547	HE22	0.99975600	53.471
14	−366.690835026	7.000000000	FK5	1.50393369	53.342

TABLE 6-continued

15	168.903118236	22.511779576	HE22	0.99975600	53.644
16	−139.218031623	8.985080714	FK5	1.50393369	54.208
17	−123.567400000	2.500000000	HE22	0.99975600	56.004
18	−128.783147634	13.174515048	FK5	1.50393369	56.475
19	277.363638285	16.146941688	HE22	0.99975600	65.608
20	−374.603552926	11.576860915	SIO2HL	1.47449062	67.030
21	−343.089730000AS	0.711044489	HE22	0.99975600	71.729
22	−420.199838469	18.542914675	FK5	1.50393369	72.916
23	−155.539539836	17.496701961	HE22	0.99975600	75.336
24	1333.809093290	36.124714972	FK5	1.50393369	91.010
25	−185.832172282	0.700000000	HE22	0.99975600	92.832
26	250.721900038	41.103381343	FK5	1.50393369	95.351
27	−332.445610329	0.700000000	HE22	0.99975600	94.503
28	116.444713685	47.036743919	FK5	1.50393369	79.545
29	111.619277578	16.476537641	HE22	0.99975600	62.131
30	447.447071546	7.025570125	LLF1	1.57916392	61.355
31	129.972246168	6.762847605	HE22	0.99975600	56.413
32	209.060715792	6.000000000	LLF1	1.57916392	56.067
33	142.896213121	23.909037803	HE22	0.99975600	53.895
34	−143.341221627	6.000000000	LLF1	1.57916392	53.429
35	229.232144802	26.400382259	HE22	0.99975600	54.656
36	−102.036060134	6.945363072	LLF1	1.57916392	55.469
37	657.537737065	12.681324161	HE22	0.99975600	65.553
38	−530.143894386	27.703305073	FK5	1.50393369	69.472
39	−159.801566428	0.700000000	HE22	0.99975600	76.285
40	765.615799963	32.449421705	FK5	1.50393369	89.488
41	−230.219500740	25.661035204	HE22	0.99975600	91.709
42	−2006.326386923	23.268749957	FK5	1.50393369	100.085
43	−281.880872075	7.000000000	HE22	0.99975600	101.130
44	0.000000000	0.000000000	HE22	0.99975600	103.328
45	812.988201022	25.496187393	FK5	1.50393369	106.335
46	−483.887830132	0.700000000	HE22	0.99975600	106.775
47	409.386219611	30.113836534	FK5	1.50393369	106.429
48	−620.224929129	13.212599489	HE22	0.99975600	105.746
49	−277.990919522	10.000000000	LLF1	1.57916392	105.366
50	264.328541579	10.759380926	HE22	0.99975600	105.744
51	259.384813207	56.002902356	FK5	1.50393369	109.552
52	−293.158581481	0.700000000	HE22	0.99975600	109.747
53	187.834974740	29.248655554	FK5	1.50393369	100.299
54	817.293359840	0.700000000	HE22	0.99975600	98.298
55	147.554780891	37.816848863	FK5	1.50393369	87.855
56	−1882.512255332	4.274618665	HE22	0.99975600	84.162
57	−689.155322670	8.806734917	LLF1	1.57916392	82.907
58	94.754605161	0.700000000	HE22	0.99975600	65.311
59	91.187562936	50.682577356	SIO2HL	1.47449062	64.971
60	130.370092405	0.700941997	HE22	0.99975600	50.836
61	82.829181894	56.332404098	SIO2HL	1.47449062	48.357
62	1067.571768894	1.722932437	HE22	0.99975600	27.548
63	0.000000000	5.000000000	SIO2HL	1.47449062	26.373
64	0.000000000	12.000000000	L710	0.99998200	23.910
65	0.000000000	0.000320000	L710	0.99998200	13.603
66	0.000000000	0.000000000	L710	0.99998200	13.603

ASPHERICAL CONSTANTS
SURFACE NO. 21

K	10.1373
C1	4.01395910e−008
C2	1.18134520e−013
C3	4.52594060e−017
C4	−2.95946970e−021
C5	1.11740260e−024
C6	−6.29238120e−029

An aspherical surface which is marked in the tables in the second column, in addition, by the abbreviation "AS" is described by the asphere formula:

$$p(h) = \frac{\frac{h^2}{r}}{1+\sqrt{1-(1+K)\frac{h^2}{r^2}}} + C1\cdot h^4 + C2\cdot h^6 + \ldots$$

wherein

r is a radius in the vertex of the asphere

h is a distance of a surface point from the optical axis,

p(h) is a arrow height, that is, the distance from a planar surface traversing the surface vertex and is oriented perpendicularly to the optical axis,

K is a conical constant and

C1 . . . C6 are aspherical constants.

Except for the lens surface **21** of lens **10** disposed towards the second object **105**, all lens surfaces are spherical lens surfaces. The classification of this three-bulge system into the five lens groups is as follows:

Lenses **1** to **4** are allocated to a first lens group G_1 having a positive refractive power. Lenses **5** to **9** are allocated to a

second lens group G_2 having a negative refractive power. Lenses **10** to **14** are allocated to a third lens group G_3 having, as a whole, a positive refractive power. Lenses **15** to **18** are allocated to a fourth lens group G_4 having, as a whole, a negative refractive power. Lenses **19** to **30**, plus lens **31**, are allocated to a fifth lens group G_5 having, as a whole, a positive refractive power.

Merely lenses **15**, **16**, **17**, **18**, **24** and **28** are made of materials of the second material group. All of these lenses are made of the LLF1 material. Accordingly, this objective comprises no lens at all which is made of the LF5 material which, of the materials of the second material group, namely LLF1 and LF5, is more sensitive to radiation.

The fifth lens group G_5 merely includes two lenses of flint glass, namely lens **24** having an almost maximal free diameter and lens **28**. Even this lens **28** still has a relatively large free diameter which is 0.765 times the largest free diameter of lens **25**.

The terminal plate **31** is made of quartz glass and has a thickness of 5 mm and can thus contribute to a considerable extent to the correction of an image field curvature. Lenses **29** and **30** contiguous to this plane-parallel plate **31** are particularly thick lenses of quartz glass. Lens **29** has a ratio of thickness to free diameter of 0.388. For lens **30** this ratio is 0.579. Of the first to fourth lens groups, merely the fourth group G_4comprises lenses made of the LLF1 material, that is, a material of the second material group. The lens groups G_1, G_2 and G_3 are completely free of lenses made of the second material group.

This particularly refined design is also enabled, among others, due to the fact that one lens, namely lens **10**, has an aspherical surface. This aspherical surface enables corrections which, on the other hand, allow to reduce the lens volume of materials of the second material group. Here, lens **10** is the marginal lens of the third lens group G_3 which delimits this lens group G_3 from the lens group G_2. In the region of lens **10** the beam path turns from the first waist into the second bulge and, as is evident from FIG. **7**, in particular, the curvature of the lower marginal beam **119** of the lower outermost field point **121** exhibits a curvature deflection point.

An objective **101** shown in FIG. **8** exhibits the optical data indicated in Table 7 below:

TABLE 7

Surface	Radii	Thicknesses	Medium	Refractive index	½ free diameter
0	0.000000000	32.000000000	L710	0.99998200	54.410
1	0.000000000	0.750000000	L710	0.99998200	61.227
2	257.086302851AS	27.456447389	FK5	1.50393369	62.787
3	−209.777767013	5.733033666	L710	0.99998200	63.759
4	2246.853389192	23.793787473	FK5	1.50393369	63.327
5	−138.529935896	11.758456054	L710	0.99998200	63.121
6	−146.562707269AS	7.000000000	FK5	1.50393369	56.832
7	150.198866750	19.695432873	L710	0.99998200	53.218
8	−161.550278545	7.000000000	FK5	1.50393369	53.193
9	230.213267801	21.668229260	L710	0.99998200	55.439
10	−135.020922361	7.000000000	FK5	1.50393369	56.210
11	275.793524158	31.866782379	L710	0.99998200	65.618
12	−720.783416016	21.068987402	FK5	1.50393369	82.294
13	−197.724036374	0.750000000	L710	0.99998200	85.699
14	−2932.235141013	32.346252327	FK5	1.50393369	96.526
15	−202.687476756	0.750000000	L710	0.99998200	99.423
16	556.989011171	44.824840928	FK5	1.50393369	109.885
17	−261.299354979	0.750000000	L710	0.99998200	110.868
18	168.067932476	43.818178326	FK5	1.50393369	106.214
19	2212.229044278	0.750000000	L710	0.99998200	103.671
20	154.602221691	27.974537641	FK5	1.50393369	91.097
21	511.109614775AS	1.427020587	L710	0.99998200	87.028
22	203.750042471	10.957454904	FK5	1.50393369	80.716
23	97.637022780	27.072757658	L710	0.99998200	67.572
24	912.104284745	7.000000000	LLF1	1.57916392	66.476
25	129.851178911	29.850899965	L710	0.99998200	60.422
26	−157.865355431	7.000000000	LLF1	1.57916392	59.802
27	310.030976716	30.535116324	L710	0.99998200	60.628
28	−97.189133534	16.225403180	LLF1	1.57916392	61.134
29	978.355427585	11.131382680	L710	0.99998200	76.843
30	−465.646326964	32.696102030	FK5	1.50393369	78.047
31	−175.840701401	1.107688422	L710	0.99998200	86.842
32	−1255.602600202	30.875369801	FK5	1.50393369	97.446
33	−206.718714998	0.750000000	L710	0.99998200	100.506
34	625.328941866	55.961745005	FK5	1.50393369	110.152
35	−195.406107120	4.758523121	L710	0.99998200	111.367
36	−179.946479842	12.000000000	LF5	1.61906768	111.129
37	−326.560052532	6.600000000	L710	0.99998200	116.310
38	0.000000000	1.000000000	L710	0.99998200	117.931
39	880.277431480	37.591715990	FK5	1.50393369	119.769
40	−386.537563150	0.750000000	L710	0.99998200	120.610
41	−387.452099603	12.000000000	LF5	1.61906768	120.578
42	−486.582839626	0.750000000	L710	0.99998200	122.142
43	338.819180262	55.758188246	FK5	1.50393369	122.238
44	−353.677034290	0.942122740	L710	0.99998200	120.766
45	−351.560711397	12.000000000	LF5	1.61906768	120.353
46	−2553.305133800	0.750000000	L710	0.99998200	117.638

TABLE 7-continued

47	278.431299377	9.999998390	LF5	1.61906768	112.851
48	141.274492122	1.449051032	L710	0.99998200	104.112
49	142.560885188	56.685027449	FK5	1.50393369	104.345
50	4205.079723653	0.750000000	L710	0.99998200	101.627
51	125.239909170	24.638559390	FK5	1.50393369	89.597
52	211.511418953	0.750000000	L710	0.99998200	86.719
53	129.722868649	31.499453310	FK5	1.50393369	80.852
54	933.062064234	4.174005465	L710	0.99998200	76.791
55	0.000000000	40.807448591	FK5	1.50393369	75.626
56	0.000000000	3.000000000	L710	0.99998200	51.890
57	0.000000000	40.000000000	FK5	1.50393369	48.423
58	0.000000000	10.000000000	L710	0.99998200	25.157
59	0.000000000		L710	0.99998200	13.602

ASPHERICAL CONSTANTS
SURFACE NO. 2

K	0.0000
C1	−1.56147159e−008
C2	−1.52606461e−011
C3	−2.08874252e−016
C4	−3.92787668e−020
C5	1.72279373e−023
C6	−5.82399558e−028

ASPHERICAL CONSTANTS
SURFACE NO. 6

K	0.0000
C1	2.20837511e−007
C2	1.12020181e−011
C3	1.81576327e−016
C4	1.17156759e−019
C5	−6.17971012e−024
C6	−9.86320452e−028

ASPHERICAL CONSTANTS
SURFACE NO. 21

K	0.0000
C1	3.89758560e−008
C2	−1.18276360e−012
C3	6.47133946e−017
C4	−5.50034606e−021
C5	3.40396458e−025
C6	−1.66626084e−029

Except for the lens surfaces **2** and **6** of the lenses **1** and **3**, respectively, disposed towards the first object **103** and the lens surface **21** of lens **10** disposed towards the second object **105**, all surfaces are spherical surfaces.

The classification of this three-bulge system into the five groups is as follows:

Lenses **1** to **2** are allocated to a first lens group G_1 having a positive refractive power. Lenses **3** to **5** are allocated to a second lens group G_2 having a negative refractive power. Lenses **6** to **10** are allocated to a third lens group G_3 having, as a whole, a positive refractive power. Lenses **11** to **15** are allocated to a fourth lens group G_4 having, as a whole, a negative refractive power. Lenses **16** to **28** are allocated to a fifth lens group G_5 having, as a whole, a positive refractive power.

Here, a partial group **131** consisting of the five lenses with numbers **24** to **28** is made of the FK5 material. Accordingly, the five lenses disposed closest to the second object **105** or wafer are not made of the material of the second material group which has a relatively higher sensitivity to radiation. The first lens, viewed from the wafer, which is made of the LF5 material, a material of the second material group, is lens **23**. This lens has already a nearly maximum free diameter, namely 92.3% of the free diameter of the largest lens number **20**. Moreover, lenses **22** and **19** of the fifth lens group G_5 are made of the LF5 material, said lenses **19** and **22** having free diameters which are about as large as that of lens **23**. Accordingly, all lenses of the fifth lens group which are made of the second material group are very large lenses and thus subjected to a relatively low radiation density. Moreover, these lenses are also transilluminated particularly uniformly because in the cross-section thereof the beam bundles which emanate from the diametrically opposed outermost field points have a little displacement q of maximally 3% of the bundle diameter (see FIG. **3**).

Besides the fifth lens group G_5, merely the fourth lens group G_4, that is, the second beam waist, comprises lenses made of materials of the second material group. The lens groups G_1, G_2 and G_3 are completely free of the materials of the second material group. All lenses of these lens groups G_1, G_2 and G_3 are made of the FK5 material.

This very advantageous configuration of the system **101** is enabled, among others, due to the fact that by the lenses of groups G_1, G_2 and G_3 a small number of aspherical surfaces is provided. The lenses **1**, **3** and **10** each have one aspherical surface. Here, lenses **3** and **10** are each marginal lenses of their lens groups G_2 and G_3, respectively, wherein lens **3** is the marginal lens of the second lens group G_2 disposed

towards the lens group G_1, and lens **10** is the marginal lens of the lens group G_3 disposed towards the lens group G_4. Moreover, the surfaces of the marginal lenses (numbers **3**, **10**) disposed away from the respective lens groups G_2 and G_3 are aspherical surfaces. Accordingly, these are the surfaces of the marginal lenses where a deflection point can most likely be allocated to the beam path. In these regions of the objective, the possibility to correct optical errors, e.g., chromatic errors is particularly high and thus the effect of aspherical surfaces is particularly strong.

The configuration which is particularly favorable as regards the stability of the optical properties provided by the system **101** is, among others, achieved in that the third bulge of the beam path, that is, the fifth lens group G_5, comprises three pseudo-cemented lens pairs. These are the lens pairs formed of lenses **18** and **19**, **21** and **22**, and **23** and **24**, respectively. One lens of each of these lens pairs, namely lenses **18**, **21** and **24** are made of the FK5 material and have a positive refractive power, whereas the respective other lenses of the lens pair, namely lenses **19**, **22** and **23** are made of the LF5 material and have a negative refractive power. Moreover, the lens surfaces disposed towards each other of these pairs have radii of curvature which deviate only slightly from each other, and the distances of these lens surfaces disposed towards each other are relatively small. In particular, in the pseudo-cemented lens pairs, the lenses having a negative refractive power (**19**, **22** and **23**) are convex-concave lenses.

Moreover, the pseudo-cemented lens pair **18**, **19** is disposed in the beam path on the left side of a system diaphragm **123**, that is, closer to the first object. The lens surfaces disposed towards each other of this pseudo-cemented lens pair both have a center of curvature which, in regard to the lens pair, is disposed closer to the first object. This means that these two lens surfaces are bulged towards the second object. Vice versa, the surfaces **48**, **49** of the pseudo-cemented lens pair formed of lenses **23** and **24** are bulged towards the first object, that is, towards the mask. Accordingly, the centers of curvature thereof are, in regard to the lenses, disposed on the side of the second object.

The lenses (**28** and **27**) disposed closest to the second object are each provided as plane-parallel plates. The thicknesses thereof are relatively large. For lens **28** a ratio of thickness to free diameter is 0.413 and for lens **27** 0.291.

A further embodiment of the projection exposure system **101** is shown in FIG. **9** and has the optical data indicated in Table 8 below:

TABLE 8

Surface	Radii	Thicknesses	Medium	Refractive index	½ free diameter
0	0.000000000	32.000000000	L710	0.99998200	58.140
1	0.000000000	0.750000000	L710	0.99998200	65.189
2	282.192219894AS	27.457052176	FK5	1.50393369	66.895
3	−241.381215170	1.136949390	L710	0.99998200	67.889
4	405.519556468	28.572155428	FK5	1.50393369	67.776
5	−162.327564008	8.858286438	L710	0.99998200	67.270
6	−179.515682000AS	7.000000000	FK5	1.50393369	61.430
87	118.671268436	26.564134389	L710	0.99998200	56.194
8	−134.683121024	7.000000000	FK5	1.50393369	56.193
9	148.266383499	28.702061130	L710	0.99998200	60.848
10	−174.097992291	7.000000000	FK5	1.50393369	63.018
11	485.623450869AS	14.345038406	L710	0.99998200	74.336
12	−687.547260677	23.501456012	FK5	1.50393369	80.224
13	−174.368626089	0.750000000	L710	0.99998200	84.135
14	5893.679253667	36.091386395	FK5	1.50393369	98.532
15	−205.521224318	0.750000000	L710	0.99998200	101.609
16	568.218064064	38.110079014	FK5	1.50393369	112.198
17	−354.923718901	0.750000000	L710	0.99998200	113.078
18	280.594221879	37.712148355	FK5	1.50393369	112.631
19	−1057.381915182	0.750000000	L710	0.99998200	111.391
20	147.033621246	46.010425948	FK5	1.50393369	98.437
21	−11345.940350233AS	0.750000000	L710	0.99998200	93.635
22	200.357918671	12.478706933	FK5	1.50393369	81.665
23	97.768502374	33.133326081	L710	0.99998200	67.606
24	−701.580361435	7.000000000	LLF1	1.57916392	66.184
25	144.978610318	26.147037190	L710	0.99998200	60.339
26	−181.559790041	7.000000000	LLF1	1.57916392	59.950
27	266.979189364	30.790963476	L710	0.99998200	61.002
28	−101.856287378	7.000000000	LLF1	1.57916392	61.636
29	−762.628295888	3.854117064	L710	0.99998200	71.317
30	−416.729125431	7.000000000	FK5	1.50393369	71.899
31	574.980608014	8.088102241	L710	0.99998200	80.369
32	−3312.164163035	34.055164886	FK5	1.50393369	81.811
33	−160.925893173	0.750000000	L710	0.99998200	87.034
34	−2523.893889452	29.749231922	FK5	1.50393369	99.345
35	−236.956196188	0.750000000	L710	0.99998200	102.171
36	1275.048030565	48.695032094	FK5	1.50393369	109.632
37	−201.292310046	0.750000000	L710	0.99998200	111.098
38	−212.255993970	12.000000000	LF5	1.61906768	110.727
39	−415.796881796	12.600000000	L710	0.99998200	114.933
40	0.000000000	−5.000000000	L710	0.99998200	117.130
41	600.920761675	32.924992407	FK5	1.50393369	118.714
42	−631.513980140	0.750000000	L710	0.99998200	119.491

TABLE 8-continued

43	336.754523305	56.047480672	FK5	1.50393369	120.323
44	−333.201559280	0.750000000	L710	0.99998200	119.014
45	−338.496328889	12.000000000	LF5	1.61906768	118.464
46	2065.431059856	0.750000000	L710	0.99998200	115.611
47	442.741290839	10.000000000	LF5	1.61906768	114.462
48	166.804899968	3.042467082	L710	0.99998200	108.876
49	171.848735171	59.085369437	FK5	1.50393369	109.527
50	−933.935513869	0.750000000	L710	0.99998200	108.228
51	169.262017611	29.393250487	FK5	1.50393369	99.879
52	493.337795039	0.750000000	L710	0.99998200	97.471
53	115.491172564	40.249706496	FK5	1.50393369	84.565
54	983.978746971	4.288688991	L710	0.99998200	80.296
55	0.000000000	44.815189458	FK5	1.50393369	79.126
56	0.000000000	3.000000000	L710	0.99998200	52.977
57	0.000000000	40.000000000	FK5	1.50393369	49.492
58	0.000000000	10.000000000	L710	0.99998200	26.153
59	0.000000000				14.535

ASPHERIC CONSTANTS
SURFACE NO. 2

K	0.0000
C1	9.37042284e−009
C2	−1.34042308e−011
C3	−2.93763572e−016
C4	4.67776977e−020
C5	−3.20938884e−024
C6	6.45491231e−028
C7	0.00000000e+000
C8	0.00000000e+000
C9	0.00000000e+000

ASPHERIC CONSTANTS
SURFACE NO. 6

K	0.0000
C1	1.89997373e−007
C2	5.44778562e−012
C3	4.54051764e−016
C4	1.82924121e−020
C5	1.16522467e−023
C6	−2.06950311e−027

ASPHERIC CONSTANTS
SURFACE NO. 11

K	0.0000
C1	−6.63292769e−008
C2	0.00000000e+000
C3	8.79403482e−017
C4	−2.95855443e−020
C5	3.73744864e−024
C6	−3.39583572e−028

ASPHERIC CONSTANTS
SURFACE NO. 21

K	0.0000
C1	4.06377005e−008
C2	−1.49057229e−012
C3	7.25253894e−017
C4	−5.17120726e−021
C5	3.05668240e−025
C6	−8.62404836e−030

A first lens group G_1 with lenses **1** and **2** has a positive refractive power. A lens group G_2 with lenses **3** to **5** has a negative refractive power. A lens group G_3 with lenses **6** to **10** has a positive refractive power. A lens group G_4 with lenses **11** to **15** has a negative refractive power and a lens group G_5 with lenses **16** to **28** has, as a whole, a positive refractive power.

The projection exposure system **101** shown in FIG. **9** is very similar to the one shown in FIG. **8**. A main difference to the above-described objective resides in lens **5** which in this case has an aspherical surface. The lens system thus comprises a total of four lenses having an aspherical surface, one of which is included in the first lens group G_1, two in the second lens group G_2 and a fourth one in the third lens group G_3. Again, the aspherical surfaces are each disposed on a surface of a marginal lens of the respective lens group disposed towards an adjacent lens group. For example, now lens **5** belonging to the second lens group G_2 has an aspherical surface on its side disposed towards the lens group G_3.

FIG. **10** shows a further embodiment of a projection exposure system **101**. The optical data thereof are indicated in Table 9 below:

TABLE 9

Surface	Radii	Thicknesses	Medium	Refractive index	½ free diameter
0	0.000000000	32.000000000	L710	0.99998200	54.410
1	0.000000000	9.331875137	L710	0.99998200	60.836
2	340.551527021AS	39.715891248	FK5	1.50393369	64.330
3	−186.309929743	0.750000000	L710	0.99998200	64.787
4	141.542483671	33.383357176	FK5	1.50393369	60.648
5	−163.899893068	2.610573086	L710	0.99998200	58.302
6	−123.871118429AS	7.000000000	FK5	1.50393369	58.176
7	95.611801386	22.280386062	L710	0.99998200	50.403
8	−250.168881171	7.000000000	FK5	1.50393369	50.291
9	133.629519296	24.530065417	L710	0.99998200	51.234
10	−128.592330546	7.000000000	FK5	1.50393369	52.215
11	435.996432153	13.632837749	L710	0.99998200	59.973
12	−327.496289890	13.886025688	FK5	1.50393369	62.448
13	−155.059472033	0.750000000	L710	0.99998200	65.245
14	−440.826527340	15.677208019	FK5	1.50393369	70.299
15	−179.925282435	0.750000000	L710	0.99998200	72.668
16	341.908495115	33.200091325	FK5	1.50393369	82.412
17	−255.826237407	0.750000000	L710	0.99998200	83.699
18	178.301539478	33.239991729	FK5	1.50393369	85.251
19	−1226.293037186	0.750000000	L710	0.99998200	83.950
20	194.943899883	28.359618223	FK5	1.50393369	78.544
21	−749.005002551AS	0.750000000	L710	0.99998200	75.527
22	271.050782514	7.000000000	FK5	1.50393369	68.810
23	105.634522505	28.202734052	L710	0.99998200	60.981
24	−306.029742311	7.000000000	FK5	1.50393369	60.027
25	150.004855021	22.462721757	L710	0.99998200	56.690
26	−187.329555540	7.000000000	FK5	1.50393369	56.673
27	304.969854455	35.116466032	L710	0.99998200	58.843
28	−81.033917535	7.000000000	FK5	1.50393369	59.679
29	−2073.312944549	6.385902439	L710	0.99998200	75.490
30	−442.663004679	32.545158129	FK5	1.50393369	76.435
31	−134.819767610	0.750000000	L710	0.99998200	82.632
32	−1759.111734795	40.360645854	FK5	1.50393369	95.217
33	−162.361009862	0.750000000	L710	0.99998200	98.341
34	−18055.541517736	48.544948355	FK5	1.50393369	102.515
35	−158.866051572	1.645215163	L710	0.99998200	103.405
36	−156.407134239	12.000000000	LF5	1.61906768	102.914
37	−6393.529232283	6.600000000	L710	0.99998200	109.938
38	0.000000000	1.000000000	L710	0.99998200	111.602
39	817.305563716	54.187957856	FK5	1.50393369	115.189
40	−220.943401929	2.092015373	L710	0.99998200	117.550
41	−214.391627564	12.000000000	LF5	1.61906768	117.539
42	−436.013733754	0.750000000	L710	0.99998200	124.264
43	481.940058078	41.665592502	FK5	1.50393369	131.389
44	−670.182657522	0.750000000	L710	0.99998200	131.671
45	409.406477442	12.000000000	LF5	1.61906768	129.565
46	187.024832050	1.409033148	L710	0.99998200	123.455
47	188.788044323	58.072022509	FK5	1.50393369	123.830
48	10342.156208557	0.750000000	L710	0.99998200	122.453
49	261.200133621	27.412944131	FK5	1.50393369	117.932
50	769.684170935	0.750000000	L710	0.99998200	115.756
51	189.162055169	24.892089316	FK5	1.50393369	107.283
52	349.166399485	0.750000000	L710	0.99998200	104.008
53	162.346317469	54.868508820	FK5	1.50393369	95.696
54	914.235997056	4.515179029	L710	0.99998200	79.846
55	0.000000000	94.622944676	FK5	1.50393369	78.827
56	0.000000000	0.800000000	L710	0.99998200	28.498
57	0.000000000	4.000000000	FK5	1.50393369	27.700
58	0.000000000	12.000000000	L710	0.99998200	25.572
59	0.000000000				13.604

ASPHERICAL CONSTANTS
SURFACE NO. 2

K	0.0000
C1	1.27733171e−007
C2	−5.00437170e−012
C3	3.30032423e−016
C4	9.45747416e−021
C5	−8.90830573e−024
C6	2.19646028e−027

ASPHERICAL CONSTANTS
SURFACE NO. 6

K	0.0000
C1	1.18195961e−007

TABLE 9-continued

C2	3.80599397e-012
C3	-2.47232851e-015
C4	5.61195650e-019
C5	-8.66028590e-023
C6	6.89083138e-027
ASPHERICAL CONSTANTS SURFACE NO. 21	
K	0.0000
C1	4.49761819e-008
C2	-1.61311907e-012
C3	2.03519674e-016
C4	-2.44484189e-020
C5	2.03153768e-024
C6	-6.76960872e-029

In this embodiment of the projection exposure system, the first, third and fifth lens groups G_1, G_3 and G_5 with lenses **1** and **2**, **6** to **10** and **15** to **27** (plus **28**), respectively, have, as a whole, a positive refractive power. The lens groups G_2 and G_4 comprise lenses **3** to **5** and **11** to **14**, respectively, and have, as a whole, a negative refractive power.

This embodiment constitutes a particularly advantageous projection exposure system because lenses of a material of the second material group are exclusively included in the fifth lens group G_5, that is, in the third bulge. These lenses **18**, **20** and **22** are made of the LF5 glass and have each a free diameter which is larger than 80% of the diameter of the largest lens **21**. Accordingly, in this embodiment, lenses made of a material of the second material group exhibiting a higher radiation sensitivity are disposed exclusively in regions in which there is both a relative low radiation density due to the large beam bundle diameter and a homogeneous illumination of the entire lens surface due to a small displacement between different beam bundles emanating from different points of the first object plane. Accordingly, this projection exposure system **101** is relatively insensitive to radiation-induced degradation which occurs at high radiation densities.

This advantageous embodiment is enabled, among others, in that, again, one lens with an aspherical surface is disposed in each one of the lens groups G_1, G_2 and G_3 at respective advantageous marginal positions of the lens groups. Moreover, the two lenses **26** and **27** disposed close to the second object **105** are thick lenses with a ratio of thickness to free diameter of 0.311 and 0.657, respectively.

What is claimed is:

1. A projection exposure system positionable between a first object and a second object for imaging the first object in a region of the second object with light of a wavelength band having a width δλabout a central working wavelength λ, wherein a relative width δλ/λof the wavelength band is larger than 0.002, comprising:

at least a first, second, third, fourth and fifth lens group which are successively positioned in this order between the first and the second object along an optical axis and each of which comprises a plurality of lenses, wherein the first, third and fifth lens group each has a positive refractive power and the second and fourth lens group each has a negative refractive power, and wherein the fifth lens group is a lens group which is positioned closest to the second object,

wherein each lens is made of a material having an Abbe number V=(N(x)-1)/N(x-Δδ))-N(x+δλ), wherein N(x) is a refractive index of the material at a wavelength x, and Δλis a wavelength difference, and wherein the material is selected from at least two material groups, namely a first material group whose materials have Abbe numbers (V1) which are higher than a limit value (Vg), and a second material group whose materials have Abbe numbers (V2) which are lower than the limit value (Vg), wherein at least one of the lenses is made of a material of the first material group and has a positive refractive power and at least one of the lenses is made of a material of the second material group and has a negative refractive power,

wherein the fifth lens group comprises a partial group of at least three directly successively arranged lenses, wherein the partial group comprises a terminal lens which, of all lenses of the system, is the one which is disposed closest to the second object, and wherein a lens directly adjacent to the partial group is made of a material of the second material group, and wherein all lenses of the partial group are made of materials which are included in the first material group.

2. The projection exposure system according to claim **1**, wherein the partial group comprises four lenses.

3. The projection exposure system according to claim **1**, wherein the partial group comprises five lenses.

4. The projection exposure system according to claim **1**, wherein the partial group comprises at least six lenses.

5. The projection exposure system according to claim **1**, wherein a lens volume of all lenses made of materials of the second material group is smaller than 20%, preferably, smaller than 15% of a lens volume of all lenses made of materials of the first material group.

6. The projection exposure system according to claim **1**, wherein the lens made of the material of the second material group and positioned directly adjacent to the partial group has a free diameter D for which it applies: $D>0.7 * D_{max}$, preferably, $D>0.8 * D_{max}$ and, more preferred, $D>0.9 * D_{max}$, wherein D_{max} is a free diameter of a lens having the largest free diameter of all lenses.

7. The projection exposure system according to claim **1**, wherein all lenses of the fifth lens group made of materials of the second material group are positioned along the optical axis in a region wherein a first bundle cross-section of a beam bundle emanating from a first outermost field point of the first object is displaced transversely to the optical axis in respect to a second bundle cross-section of a beam bundle emanating from a second outermost field point of the first object positioned diametrically opposed to the first outermost field point by an amount q, so that it applies: $q/d<0.1$, wherein d is a free diameter of the respective lens.

8. The projection exposure system according to claim **5**, wherein all lenses of the fifth lens group which are made of

materials of the second material group and disposed between a system diaphragm and the second object are disposed in a region along the optical axis wherein q/d ±0.05.

9. The projection exposure system according to claim **1**, wherein at least one lens of the partial group has a maximum thickness along the optical axis and a free diameter such that a ratio of maximum thickness divided by diameter is larger than 0.3.

10. The projection exposure system according to claim **1**, wherein the materials of the first material group have a refractive index which is in a range of from 0.93 * NFK5(λ) to 1.03 * NFK5(λ), wherein NFK5(λ) is a refractive index of a FK5 glass of Schott at the working wavelength.

11. The projection exposure system according to claim **1**, wherein the materials of the first material group have a refractive index at a wavelength of 365.5 nm which is smaller than 1.55.

12. The projection exposure system according to claim **1**, wherein the materials of the second material group have a refractive index which is in a range of from 0.97 * NLLF1 (λ) to 1.07 * NLLF1(λ), wherein NLLF1(λ) is a refractive index of a LLF1 glass of Schott at the working wavelength.

13. The projection exposure system according to claim **1**, wherein the materials of the second material group have a refractive index at a wavelength of 365.5 nm which is higher than 1.54.

14. The projection exposure system according to claim **1**, wherein the materials of the second material group have a refractive index which is smaller than 0.99 NLF5(λ), wherein NLF5(λ) is a refractive index of a LF5 glass of Schott at the working wavelength.

15. The projection exposure system according to claim **1**, wherein the materials of the second material group have a refractive index at a wavelength of 365.5 nm which is smaller than 1.64.

16. The projection exposure system according to claim **1**, wherein the limit value Vg is determined by the equation

$$Vg=(VFK5(\lambda)+VLF5(\lambda))/2$$

wherein VFK5(λ) is an Abbe number of a FK5 glass of Schott at the working wavelength and VLF5(λ) is an Abbe number of a LF5 glass of Schott at the working wavelength.

17. The projection exposure system according to claim **1**, wherein the limit value Vg is 400 if the Abbe number is determined at a wavelength λ=365.5 nm and a wavelength difference Δλ=3 nm.

18. The projection exposure system according to claim **1**, wherein the first material group comprises quartz glass.

19. The projection exposure system according to claim **1**, wherein a numerical aperture on the side of the second object is higher than 0.6.

20. The projection exposure system according to claim **1**, wherein a numerical aperture on the side of the second object is higher than 0.65.

21. The projection exposure system according to claim **19**, wherein a Strehl number of the system is higher than 0,95, when determined with a light spectrum of a half-width value of 2 nm on both sides of the working wavelength.

22. The projection exposure system according to claim **1**, wherein a minimum distance between opposed surfaces of each pair of directly adjacent lenses is smaller than 25 mm.

23. The projection exposure system according to claim **1**, wherein a minimum distance, averaged over all pairs of directly adjacent lenses, between opposed surfaces of the respective pair is smaller than 3 mm.

24. The projection exposure system according to claim **1**, wherein the projection exposure system is telecentric on both sides thereof.

25. The projection exposure system according to claim **1**, further comprising a diaphragm which is disposed in a region of the fifth lens group.

26. The projection exposure system according to claim **1**, wherein the light comprises a light of a mercury-I line.

27. A projection exposure system positionable between a first object and a second object for imaging the first object in a region of the second object with light of a wavelength band having a width δλabout a central working wavelength λ, wherein a relative width δλ/λof the wavelength band is larger than 0.002, comprising:

at least a first, second, third, fourth and fifth lens group which are successively positioned in this order between the first and the second object along an optical axis and each of which comprises a plurality of lenses, wherein the first third and fifth lens group each has a positive refractive power and the second and fourth lens group each has a negative refractive power, and wherein the fifth lens group is a lens group which is positioned closest to the second object,

wherein each lens is made of a material having an Abbe number V=(N(x)-1)/N(x-Δλ))-N(x+Δλ), wherein N(x) is a refractive index of the material at a wavelength x, and Δλis a wavelength difference, and wherein the material is selected from at least two material groups, namely a first material group whose materials have Abbe numbers (V1) which are higher than a lint value (Vg), and a second material group whose materials have Abbe numbers (V2) which are lower than the limit value (Vg), wherein at least one of the lenses is made of a material of the first material group and has a positive refractive power and at least one of the lenses is made of a material of the second material group and has a negative refractive power,

wherein lenses of the fourth lens group are made of materials which are included in the second material group and whose refractive index at a wavelength of 365.5 nm is smaller than 1.59.

28. A projection exposure system positionable between a first object and a second object for imaging the first object in a region of the second object with light of a wavelength band having a width δλabout a central working wavelength λ, wherein a relative width δλ/λof the wavelength band is larger than 0.002, comprising:

at least a first, second, third, fourth and fifth lens group which are successively positioned in this order between the first and the second object along an optical axis and each of which comprises a plurality of lenses, wherein the first, third and fifth lens group each has a positive refractive power and the second and fourth lens group each has a negative refractive power, and wherein the fifth lens group is a lens group which is positioned closest to the second object, wherein each lens is made of a material having an Abbe number V=(N(x)-1)/N(x-Δλ))-N(xΔλ), wherein N(x) is a refractive index of the material at a wavelength x, and Δλis a wavelength difference, and wherein the material is selected from at least two material groups, namely a first material group whose materials have Abbe numbers (V1) which are higher than a limit value (Vg), and a second material group whose materials have Abbe numbers (V2) which are lower than the limit value (Vg), wherein at least one of the lenses is made of a material of the first material group and has a positive refractive power and at least one of the lenses is made of a material of the second material group and has a negative refractive power,

wherein all lenses of at least one of the first, second, third and fourth lens group are made of materials which are

contained in the first material group, and wherein at least one lens of at least one of the first, second, third and fourth lens group comprises at least one aspherical surface.

29. The projection exposure system according to claim **28**, wherein each lens group comprises at most one lens with aspherical surface.

30. The projection exposure system according to claim **28**, wherein the at least one lens having the aspherical surface is a lens which, in the direction of the optical axis, is at least one of an outer marginal lens of the lens group, and a lens directly adjacent to the marginal lens.

31. The projection exposure system according to claim **30**, wherein the lens having the aspherical surface is the outer marginal lens of the lens group and wherein a surface thereof oriented away from the lens group is the aspherical surface.

32. The projection exposure system according to claim **28**, wherein all lenses of the first and the second and the third lens group are made of materials which are included in the first material group.

33. The projection exposure system according to claim **32**, wherein furthermore all lenses of the fourth lens group are made of materials which are included in the first material group.

34. The projection exposure system according to claim **28**, wherein, of the lenses of the first, second and third lens group, the lenses which are made of materials of the second material group account for a lens volume of less than 10% in regard to a lens volume of all lenses made of materials of the second material group.

35. A projection exposure system positionable between a first object and a second object for imaging the first object in a region of the second object with light of a wavelength band having a width δλabout a central working wavelength λ, wherein a relative width δλ/λof the wavelength band is larger than 0.002, comprising:

at least a first, second, third, fourth and fifth lens group which are successively positioned in this order between the first and the second object along an optical axis and each of which comprises a plurality of lenses, wherein the first, third and fifth lens group each has a positive refractive power and the second and fourth lens group each has a negative refractive power, and wherein the fifth lens group is a lens group which is positioned closest to the second object,

wherein each lens is made of a material having an Abbe number V=(N(x)1)/N(xΔλ))-N(xΔλ), wherein N(x) is a refractive index of the material at a wavelength x, and Δλis a wavelength difference, and wherein the material is selected from at least two material groups, namely a first material group whose materials have Abbe numbers (V1) which are higher than a limit value (Vg), and a second material group whose materials have Abbe numbers (V2) which are lower than the limit value (Vg), wherein at least one of the lenses is made of a material of the first material group and has a positive refractive power and at least one of the lenses is made of a material of the second material group and has a negative refractive power,

wherein the lens group disposed closest to the second object comprises at least one pseudo-cemented lens pair, wherein the pseudo-cemented lens pair is formed of a pair of directly adjacent lenses whose lens surfaces which are oriented towards each other are spaced apart from each other at all points in the direction of the optical axis by a distance A which is smaller than 2.5 percent of a free diameter of a larger one of the two lenses, and wherein a first lens of the lens pair is made of a material of the first material group and has a positive refractive power and a second lens of the lens pair is a lens made of a material of the second material group and has a negative refractive power.

36. The projection exposure system according to claim **35**, wherein the distance A is smaller than 1.25 percent of the free diameter of the larger one of the two lenses.

37. The projection exposure system according to claim **35**, wherein two pseudo-cemented lens pairs are disposed between a diaphragm and the second object.

38. The projection exposure system according to claim **35**, wherein the lens surfaces oriented towards each other of the pseudo-cemented lens pair positioned between the diaphragm and the second object are each bulged towards the first object.

39. The projection exposure system according to claim **35**, wherein the lens surfaces oriented towards each other of the pseudo-cemented lens pair positioned between a diaphragm and the first object are each bulged towards the second object.

* * * * *

UNITED STATES PATENT AND TRADEMARK OFFICE

CERTIFICATE OF CORRECTION

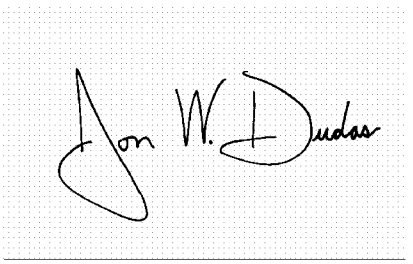
PATENT NO. : 6,806,942 B2
DATED : October 19, 2004
INVENTOR(S) : Karl-Heinz Schuster et al.

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

Column 19,
Table 5 (cont), "Thicknesses" column, delete "of a material of the Merely the second" and substitute therfor -- of a material of the second --;

Signed and Sealed this

First Day of March, 2005



JON W. DUDAS
Director of the United States Patent and Trademark Office

UNITED STATES PATENT AND TRADEMARK OFFICE

CERTIFICATE OF CORRECTION

PATENT NO. : 6,806,942 B2
DATED : October 19, 2004
INVENTOR(S) : Karl-Heinz Schuster et al.

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

Column 8,
Line 63, delete "e respective" and substitute therefor -- to the respective --;

Column 12,
Line 58, delete "20°" and substitute therefor -- 20% --;

Column 15,
Line 13, delete "in of a" and substitute therefor -- in that, of a --;

Column 18,
Line 33, delete "as a a" and substitute therefor -- as a whole, a --;

Column 19,
Line 42, delete "y the LF5" and substitute therefor -- namely the LF5 --;

Column 19, line 48 - Column 20, line 38,
Delete "of a material of the Merely the second" and substitute therfor -- of a material of the second --;

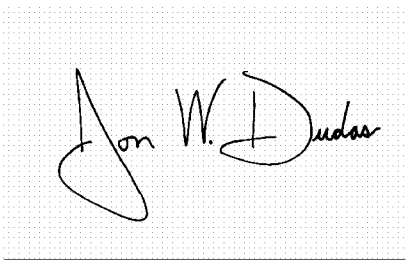
Column 20,
Lines 44-45, delete "closest wafer" and substitute therefor -- closest to the wafer --;

Column 35,
Line 3, delete "q/d $\pm$0.05" and substitute therefor -- q/d < 0.05 --; and

Column 36,
Line 26, delete "lint value" and substitute therefor -- limit value --.

Signed and Sealed this

Twenty-fourth Day of May, 2005



JON W. DUDAS
Director of the United States Patent and Trademark Office